IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 1 May 2025, accepted 19 May 2025, date of publication 22 May 2025, date of current version 12 June 2025.

*Digital Object Identifier 10.1109/ACCESS.2025.3572865*

SURVEY

# Machine Learning in E-Commerce: Trends, Applications, and Future Challenges

**ELIAS DRITSAS AND MARIA TRIGKA**
Department of Informatics and Computer Engineering, University of West Attica, 122 43 Athens, Greece

Corresponding author: Elias Dritsas (idritsas@uniwa.gr)

The publication of the article in OA mode was financially supported by HEAL-Link.

**ABSTRACT** The rapid evolution of e-commerce has been significantly influenced by the integration of machine learning (ML) and data science techniques. The present survey provides a comprehensive overview of how ML methods are applied across various functional domains in e-commerce, including personalized recommendations, dynamic pricing, fraud detection, customer segmentation, and behavioral analysis. We categorize and evaluate a wide range of ML paradigms, namely supervised, unsupervised, reinforcement, and hybrid learning, as well as emerging approaches such as neurosymbolic artificial intelligence (AI), federated learning (FL), and quantum ML (QML). Key challenges related to scalability, interpretability, cold-start problems, data sparsity, and privacy are critically analyzed. Additionally, we highlight underexplored areas, such as continual learning (CL) and multi-agent architectures in commerce. The survey incorporates comparative tables, real-world use cases, and a taxonomy of methods to support both academic and industrial perspectives. Ultimately, by analyzing trends and gaps in the literature, we provide a forward-looking research roadmap that bridges ML innovations with the evolving demands of e-commerce ecosystems.

**INDEX TERMS** Machine learning, e-commerce, predictive analytics, recommendation systems, personalization, optimization.

## I. INTRODUCTION

The rapid expansion of e-commerce has transformed how businesses interact with consumers, manage supply chains, and optimize operations. The increasing digitization of commerce has generated vast volumes of data from consumer interactions, product searches, transaction histories, and behavioural analytics. Traditional rule-based approaches to managing these data streams have proven inadequate in handling the complexity and dynamism of modern digital marketplaces. ML has emerged as a transformative force, offering intelligent automation, adaptive decision-making, and predictive analytics to enhance e-commerce efficiency [1], [2].

ML algorithms play a pivotal role in multiple aspects of e-commerce, including personalized recommendation systems, fraud detection, demand forecasting, customer segmentation, and dynamic pricing strategies. By leveraging techniques such as supervised and unsupervised learning, deep learning (DL), and reinforcement learning (RL), businesses can extract meaningful insights from data, anticipate consumer needs, and optimize operational efficiency [3].

In addition, recommendation engines, powered by collaborative filtering and neural networks, personalize user experiences by predicting product preferences. At the same time, fraud detection models employ anomaly detection techniques to secure financial transactions against cyber threats [4], [5].

Despite its transformative potential, integrating ML into e-commerce presents several challenges. Data privacy concerns, interpretability issues, computational constraints, and evolving fraud tactics pose significant obstacles to widespread adoption. Ensuring transparency in AI-driven decision-making remains a critical issue, particularly in domains such as algorithmic pricing and targeted advertising.

The associate editor coordinating the review of this manuscript and approving it for publication was Ayman El-Baz.

Furthermore, the need for scalable, real-time ML models capable of handling billions of transactions and user interactions highlights the growing demand for computational efficiency and robust model architectures [6], [7].

### A. MOTIVATION

The proliferation of e-commerce has led to increasingly complex, high-velocity digital marketplaces where consumer expectations, fraud vectors, and product dynamics evolve in real-time. In this environment, conventional rule-based systems fail to provide the adaptability, scalability, and personalization required for modern commerce. Although ML offers a path forward through intelligent automation and predictive modeling, its integration into e-commerce systems is still hindered by major challenges, including data sparsity, model interpretability, compliance constraints, and real-time processing bottlenecks.

Moreover, while the literature on ML in e-commerce is expanding, it remains fragmented across domains such as recommendation systems, fraud detection, and pricing optimization. Prior surveys often lack technical depth, focus on limited use cases, or fail to incorporate emerging paradigms like neurosymbolic AI, CL, FL, and QML. As a result, there is a pressing need for a unified, technically rigorous synthesis bridging foundational and frontier ML techniques with e-commerce-specific challenges.

This survey responds to that gap by systematically mapping ML paradigms to functional domains in digital commerce, analyzing trade-offs, and surfacing underexplored research directions. The goal is to provide researchers and practitioners with a comprehensive reference framework to guide the scalable, trustworthy, and future-proof deployment of ML in complex e-commerce ecosystems.

### B. METHODOLOGY

This survey adopts a structured and transparent methodology inspired by best practices in AI systems research, particularly the multi-stage framework outlined in recent literature on artificial general intelligence development [8]. The review process was designed to ensure both coverage and reproducibility through a sequence of defined stages, including database querying, filtering, classification, and synthesis.

We conducted a targeted search across major academic databases, such as IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, and Google Scholar. The search focused on publications from the last seven years to capture the most recent advancements in ML and its integration into digital commerce. Search queries combined terms related to core ML techniques and e-commerce domains. For example, combinations such as (“machine learning” OR “deep learning” OR “reinforcement learning”) AND “e-commerce” were used alongside more specific terms like (“recommendation systems” OR “fraud detection”) AND (“artificial intelligence” OR “machine learning”). Queries were customized per database using appropriate Boolean and truncation operators.

Inclusion criteria required that studies be peer-reviewed, written in English, and focused on the application, evaluation, or deployment of ML models within e-commerce contexts. Articles also needed to contain technical insights, system-level contributions, or architectural innovations relevant to commercial environments. We excluded purely theoretical works, non-peer-reviewed sources, non-English texts, and duplicate entries.

The selected literature was then classified across three analytical dimensions. The first considered the various types of ML approaches, including supervised, unsupervised, reinforcement, hybrid, meta-learning, neurosymbolic AI, FL, and quantum models. The second dimension focused on the functional domain, such as recommendation, customer segmentation, dynamic pricing, fraud detection, and inventory forecasting. The third dimension captured cross-cutting challenges like scalability, latency, interpretability, data sparsity, and privacy.

This thematic classification guided the taxonomy, comparative tables, and synthesis presented in subsequent sections. It also provided the basis for identifying gaps in current practice and opportunities for future research in ML-powered e-commerce systems.

Figure 1 offers a synthesized map of the survey's thematic scope, illustrating how ML techniques intersect with practical demands and unresolved challenges in e-commerce. It captures the interplay between learning strategies, their operational roles, systemic bottlenecks, and forward-looking innovations, serving as a visual reference for the structure and flow of the analysis that follows.

### C. CONTRIBUTION

This survey presents a comprehensive and forward-looking framework that not only consolidates recent ML applications in e-commerce but also extends prior work by systematically aligning ML paradigms with functional roles, system-level constraints, and emerging research challenges. In contrast to earlier surveys that either focus on narrow use cases (e.g., churn prediction or recommendation only) or offer generalized overviews without technical depth, our paper makes the following distinct contributions

- We dissect five ML paradigms—supervised, unsupervised, reinforcement, hybrid, and meta-learning, linking each to specific e-commerce functions (e.g., fraud detection, inventory management, conversational AI) and elaborating on architectural variants and computational trade-offs. This moves beyond taxonomy to a functional and operational mapping.
- The survey is among the first to contextualize neurosymbolic AI, QML, multimodal AI, and decentralized ML frameworks (e.g., FL, blockchain-based AI) within e-commerce environments. Prior surveys have largely

overlooked these paradigms or treated them in isolation from core ML applications.
- We propose a structured comparison of limitations, such as data sparsity, latency, adversarial robustness, and interpretability, and match them with targeted ML strategies (e.g., CL for model drift, graph neural networks for fraud adaptation, or edge AI for latency reduction). This enables solution-oriented guidance, filling the gap between theory and deployment.
- Section VI presents a systematic meta-analysis of various major prior surveys, highlighting how our work advances beyond them in scope (encompassing more application domains), depth (offering algorithmic insights), and foresight (providing a research roadmap). In doing so, we bridge the foundational literature with disruptive innovations, a gap absent in earlier reviews.
- The inclusion of Figure 1 and Tables 1–5 presents an integrated visual taxonomy that connects learning types, functional domains, technical challenges, and emerging research directions—serving as a reference model for both academic researchers and practitioners in digital commerce.

The remainder of the paper is structured as follows. Section II focuses on the methodologies and learning paradigms in e-commerce. Section III notes ML applications in e-commerce. Moreover, Section IV provides challenges and limitations. Besides, Section V outlines future research directions. Next, Section VI discusses the findings and compares them with existing research on ML in e-commerce. Finally, Section VII summarizes the findings of this survey.

## II. METHODOLOGIES AND LEARNING PARADIGMS

ML methodologies in e-commerce encompass a diverse set of paradigms, each designed to address distinct challenges across recommendation systems, fraud detection, inventory forecasting, pricing strategies, and customer engagement. The evolution of learning paradigms has expanded from conventional supervised and unsupervised learning to more sophisticated RL frameworks, meta-learning techniques, and hybrid models that integrate multiple approaches to optimize performance. Understanding these methodologies' computational foundations and practical implications provides deeper insights into their effectiveness in real-world e-commerce applications.

### A. SUPERVISED LEARNING FOR PREDICTIVE AND PRESCRIPTIVE ANALYTICS

Supervised learning remains one of the most extensively applied methodologies in e-commerce, offering robust predictive capabilities through labelled data-driven model training. This paradigm underpins applications such as fraud detection, customer retention prediction, and demand forecasting. Traditional approaches rely on logistic regression and decision trees. Still, recent advances incorporate ensemble methods, such as gradient boosting machines (GBM) and DL architectures like convolutional neural networks (CNNs) and recurrent neural networks (RNNs). These techniques enhance model generalization by capturing complex historical transactional and behavioural data patterns [9], [10], [11].

In fraud detection, supervised models train on labelled datasets containing historical fraud cases, enabling them to differentiate between legitimate and anomalous transactions accurately. However, the reliance on annotated data poses challenges, particularly when fraud patterns evolve dynamically. To mitigate this, semi-supervised learning techniques, which leverage small amounts of labelled data combined with a large volume of unlabeled data, have been integrated into modern fraud detection frameworks [12], [13].

Supervised learning also plays a pivotal role in customer sentiment analysis, where natural language processing (NLP) models classify user reviews and feedback into predefined categories. Transformer-based architectures, such as BERT (Bidirectional Encoder Representations from Transformers) and RoBERTa (a Robustly Optimized BERT Pretraining Approach), refine sentiment prediction by understanding contextual dependencies, significantly improving the accuracy of automated sentiment classification. In the domain of demand forecasting, long short-term memory (LSTM) networks and Transformer-based time-series models have been increasingly adopted, offering superior performance in capturing seasonality and trend shifts within sales data [14], [15], [16].

### B. UNSUPERVISED LEARNING FOR PATTERN RECOGNITION AND BEHAVIORAL CLUSTERING

Unlike supervised learning, which relies on labelled datasets, unsupervised learning discovers latent structures within data without predefined categories. This paradigm is particularly effective in customer segmentation, anomaly detection, and market trend analysis. This includes clustering algorithms such as k-means, Gaussian mixture models (GMM), and hierarchical clustering which are widely employed to identify distinct customer groups based on purchase history, browsing behaviour, and demographic attributes. The resulting segmentation allows e-commerce businesses to personalize marketing campaigns, optimize pricing strategies, and enhance user engagement by catering to specific consumer preferences [17], [18], [19].

Beyond clustering, dimensionality reduction techniques such as principal component analysis (PCA) and t-distributed stochastic neighbour embedding (t-SNE) facilitate extracting meaningful features from high-dimensional data. These methods are crucial in reducing computational overhead while preserving essential data characteristics, which is valuable for recommendation engines that require feature-rich user and product representations [20], [21].

In fraud detection, unsupervised anomaly detection models, including autoencoders and isolation forests, identify deviations from normal transaction patterns, detecting previously unseen fraudulent activities. These techniques operate

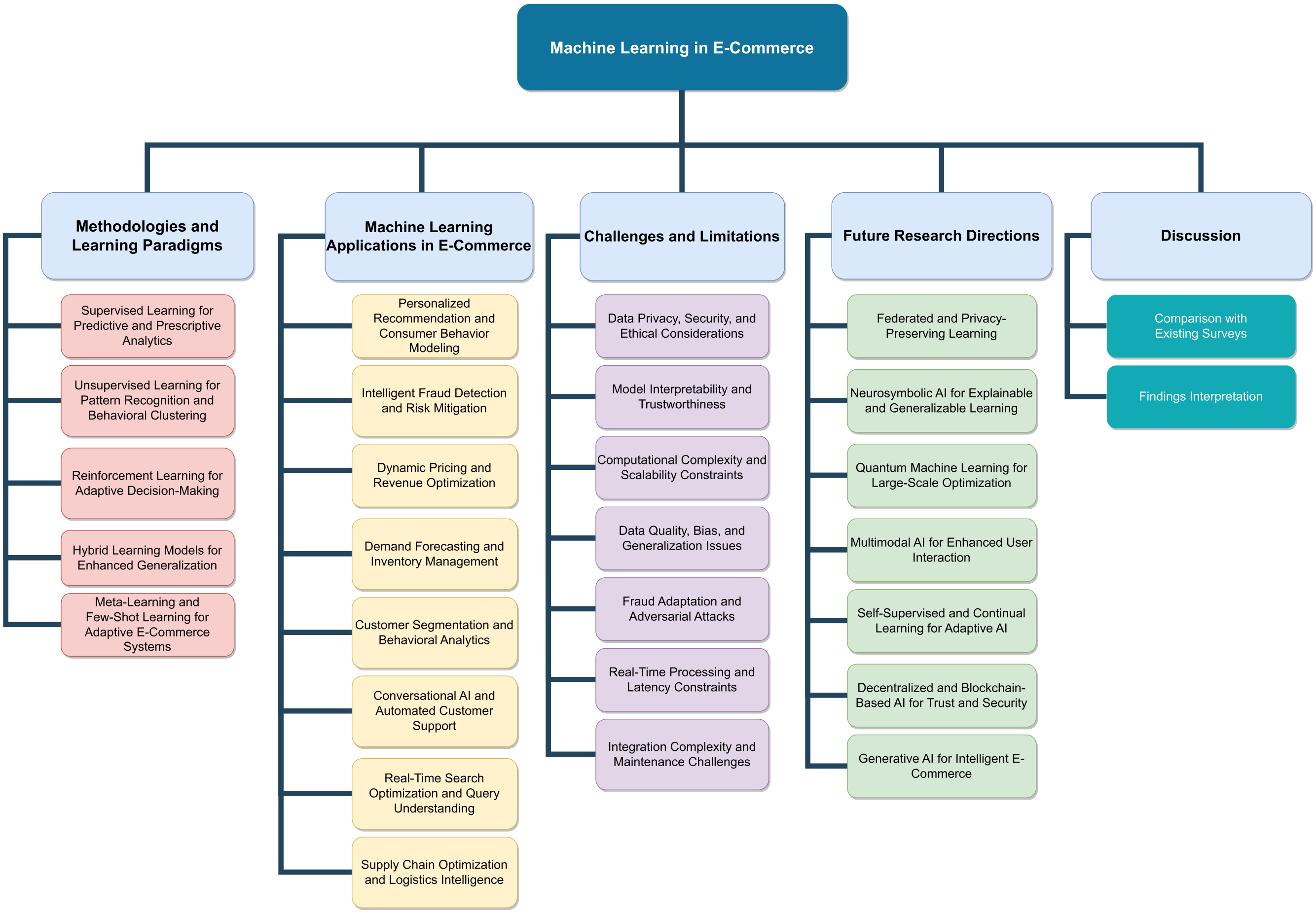



**FIGURE 1.** Surveyed topics in E-Commerce: Learning Paradigms, Functional Domains, Challenges, and Future Research Directions.

without explicit fraud labels, making them particularly advantageous in scenarios where fraudulent behaviours continuously evolve. Moreover, generative models such as variational autoencoders (VAEs) and generative adversarial networks (GANs) are increasingly utilized to generate synthetic user behaviour profiles, augmenting training datasets and improving the robustness of predictive models [22], [23], [24], [25].

### *C. REINFORCEMENT LEARNING FOR ADAPTIVE DECISION-MAKING*

RL has emerged as a powerful paradigm for optimizing dynamic decision-making processes in e-commerce. Unlike traditional learning approaches that rely on static datasets, RL continuously interacts with its environment, refining strategies based on observed rewards. This methodology has gained prominence in dynamic pricing, personalized recommendations, and automated customer support, where real-time adaptability is essential [26], [27].

In dynamic pricing, RL agents simulate market conditions, adjusting product prices in response to demand fluctuations, competitor pricing, and macroeconomic indicators. Deep RL, particularly deep Q-networks (DQNs) and policy gradient methods, enhances pricing strategies by continuously learning from transaction data, optimizing revenue generation, and maintaining customer satisfaction [28], [29].

Recommendation systems benefit significantly from RL, where policy-based learning frameworks complement traditional collaborative filtering models. Unlike static models that generate recommendations based on historical interactions, RL-powered systems adapt to real-time user feedback, optimizing engagement by presenting contextually relevant products. Multi-armed bandit algorithms, a subset of RL, balance exploration and exploitation, dynamically refining recommendation policies based on user responses [30], [31], [32].

Additionally, RL has been effectively applied to supply chain and logistics optimization, which aids in route planning, warehouse management, and delivery scheduling. RL agents optimise resource allocation by modelling logistics operations as sequential decision-making problems while minimizing operational costs and delivery times. Multi-agent RL (MARL) extends this paradigm by coordinating interactions among multiple autonomous agents, enhancing decision-making efficiency in complex supply chain networks [33], [34], [35].

### D. HYBRID LEARNING MODELS FOR ENHANCED GENERALIZATION

Hybrid learning models, which integrate multiple learning paradigms, have gained traction in e-commerce due to their ability to leverage the strengths of different approaches. These models combine supervised, unsupervised, and RL techniques to enhance generalization, robustness, and adaptability across diverse applications [36], [37].

For instance, hybrid recommendation systems integrate content-based filtering with collaborative filtering and RL to address challenges such as data sparsity and evolving user preferences. By combining explicit and implicit feedback mechanisms, these models improve recommendation accuracy and personalization [38].

In fraud detection, hybrid approaches merge supervised classification with unsupervised anomaly detection, ensuring comprehensive fraud identification even in previously unseen attack scenarios. By leveraging ensemble learning techniques such as stacking and boosting, hybrid fraud detection models significantly enhance predictive performance while reducing false positives [39], [40], [41].

Conversational AI systems also benefit from hybrid learning frameworks, where supervised fine-tuning of transformer models is combined with RL to optimize chatbot interactions. RL from human feedback (RLHF) refines chatbot responses based on user engagement metrics, continuously improving dialogue quality and contextual relevance [42], [43], [44].

Hybrid learning is further applied in demand forecasting, where supervised LSTM models are augmented with RL-based inventory control mechanisms. This fusion enables more accurate demand predictions while dynamically adjusting inventory replenishment strategies in response to real-time market conditions [45], [46].

### E. META-LEARNING AND FEW-SHOT LEARNING FOR ADAPTIVE E-COMMERCE SYSTEMS

Meta-learning, often called "learning to learn," has emerged as a transformative approach in scenarios where e-commerce systems must rapidly adapt to new data with minimal retraining. Unlike conventional ML models that require extensive labelled datasets, meta-learning frameworks generalize across tasks, enabling rapid adaptation to novel products, user behaviours, and emerging trends [47], [48], [49].

Few-shot learning, a subset of meta-learning, is particularly valuable for cold-start recommendation problems where historical data is limited. By leveraging episodic training and metric-based learning, few-shot models learn representations that generalize effectively to unseen items and users. This capability significantly enhances recommendation systems for newly launched products, ensuring personalized suggestions even in data-scarce environments [50], [51], [52].

Moreover, meta-learning is crucial in personalized search ranking, where search engines must continuously adapt to individual user preferences. Instead of retraining models from scratch, meta-learning frameworks fine-tune ranking algorithms based on limited user interactions, optimizing search relevance and user satisfaction [53], [54].

The overall landscape of ML methodologies, corresponding models, and e-commerce applications is illustrated in Figure 2. Moreover, Table 1 presents an analytical and comparative overview of core ML paradigms applied in e-commerce. It systematically contrasts five primary approaches, supervised learning, unsupervised learning, RL, hybrid learning models, and meta-learning/few-shot learning across three critical dimensions: their typical e-commerce use cases, algorithmic strengths, and context-specific limitations. By introducing concrete examples (e.g., real-time cold-start personalization or fraud detection with evolving patterns), the table supports a more nuanced understanding of the trade-offs associated with each methodology in real-world deployment.

## III. MACHINE LEARNING APPLICATIONS IN E-COMMERCE

The proliferation of ML in e-commerce has redefined how digital marketplaces operate, enhancing the efficiency and intelligence of various ecosystem components. ML's integration enables a deeper understanding of consumer behaviour, fraud mitigation, demand prediction, supply chain management, and marketing optimization. The diverse applications of ML within e-commerce encompass a spectrum of interrelated domains that collectively enhance user experience, operational effectiveness, and business intelligence.

### A. PERSONALIZED RECOMMENDATION AND CONSUMER BEHAVIOR MODELING

Modern e-commerce platforms employ sophisticated ML algorithms to predict and influence consumer purchasing patterns, creating personalized shopping experiences tailored to individual preferences. Traditional approaches relied on heuristic-based filtering techniques, but advancements in DL have introduced neural collaborative filtering and RL frameworks that dynamically adapt to users' evolving interests. ML models construct user embeddings that reflect underlying purchasing tendencies by analyzing transactional histories, search queries, browsing behaviours, and contextual cues [55], [56], [57].

Context-aware recommendation engines further refine this process by incorporating temporal signals, geographical preferences, and session-based interactions to provide real-time, highly relevant product suggestions. Initially developed for NLP, transformer-based architectures have demonstrated remarkable effectiveness in sequential recommendation tasks, capturing long-term dependencies in consumer behaviour. Hybrid models combining content-based and collaborative filtering techniques enhance robustness against data sparsity and cold-start issues, ensuring accurate predictions even for new users and products [58], [59], [60].

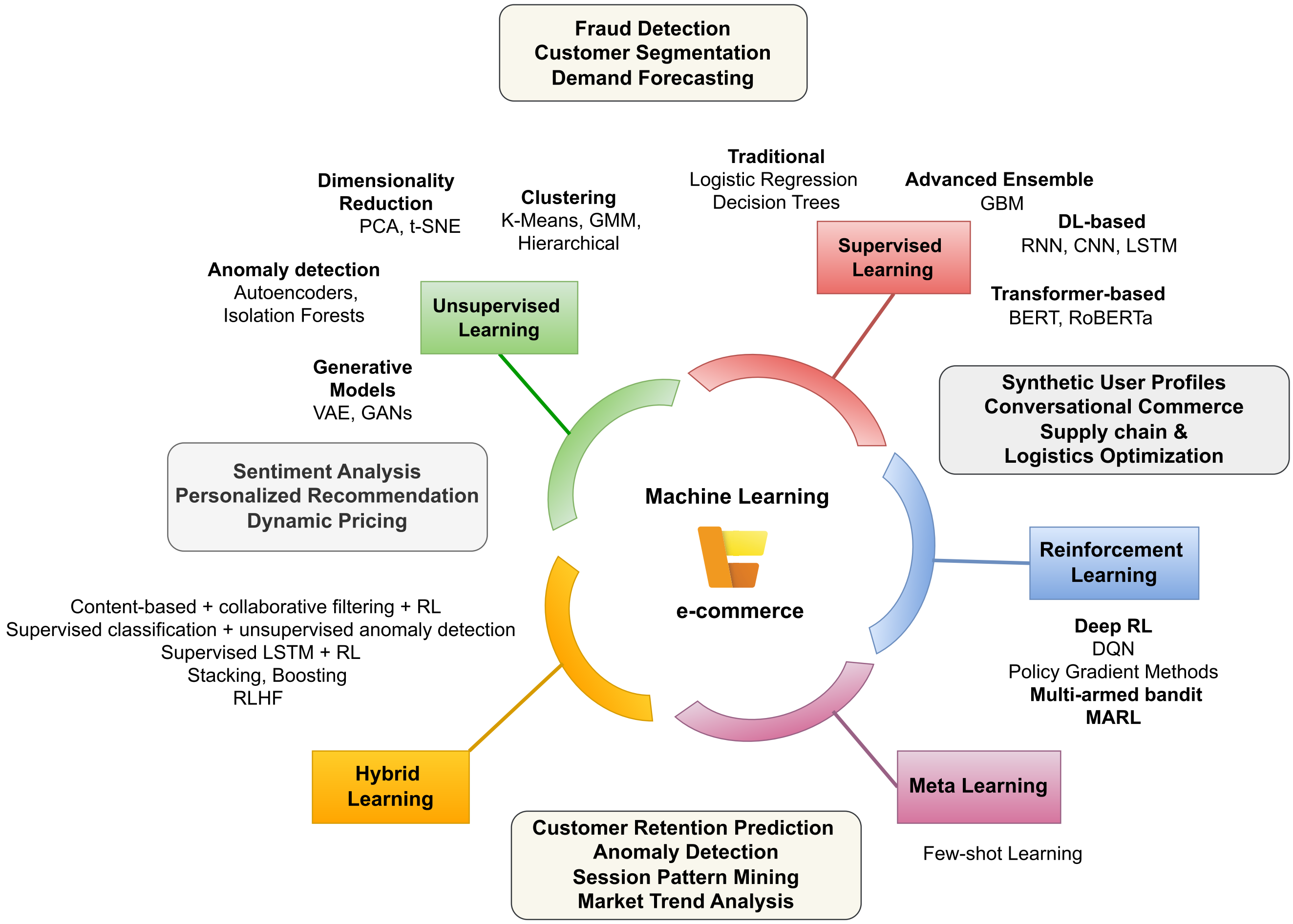



**FIGURE 2.** Landscape of Machine Learning Paradigms, Techniques, and Applications in E-Commerce.

### B. INTELLIGENT FRAUD DETECTION AND RISK MITIGATION

The expansion of online transactions has heightened concerns regarding fraudulent activities, necessitating the deployment of ML-driven security mechanisms. Unlike rule-based fraud detection systems that rely on predefined thresholds, ML models employ anomaly detection techniques to identify suspicious patterns with high precision. Supervised learning approaches, such as ensemble classifiers, leverage labelled fraud datasets to distinguish legitimate transactions from fraudulent ones. However, the dynamic nature of cyber threats necessitates adopting unsupervised and semi-supervised learning strategies to detect previously unseen fraud patterns [61], [62], [63].

Graph-based ML models have emerged as powerful tools for financial fraud detection. They analyze transactional networks and identify malicious entities based on their relational structures. These models construct a graph representation of financial activities, detecting collusion and fraudulent behaviour that traditional statistical approaches might overlook. Additionally, adversarial ML techniques are being explored to enhance fraud detection robustness by proactively identifying vulnerabilities that fraudsters may exploit. Integrating ML with blockchain-based authentication systems further strengthens security protocols, reducing identity theft and payment fraud risks [64], [65], [66], [67].

### C. DYNAMIC PRICING AND REVENUE OPTIMIZATION

Implementing ML in pricing strategies enables e-commerce platforms to generate optimal revenue while maximizing consumer satisfaction. Traditional static pricing mechanisms are replaced with dynamic pricing algorithms that adjust prices based on market demand, competitor pricing, user behaviour, and external factors such as economic indicators. RL frameworks, particularly those employing DQNs, optimize pricing decisions by continuously learning from real-time sales data and market fluctuations [68], [69].

Predictive analytics is crucial in estimating price elasticity, allowing businesses to implement personalized discounting strategies that maximize conversions. Bayesian optimization techniques further refine pricing models by balancing exploration and exploitation, enabling adaptive price adjustments that prevent revenue loss due to suboptimal pricing. Integrating sentiment analysis into pricing models helps businesses

**TABLE 1. Taxonomy of machine learning paradigms: applicability, advantages, and drawbacks in E-Commerce.**

| ML Paradigm | Use Cases in E-Commerce | Advantages | Limitations |
|---|---|---|---|
| Supervised Learning [9] [10] [11] [12] [13] [14] [15] [16] | Fraud detection, demand forecasting, churn prediction | High accuracy with labeled data; scalable for classification/regression | Requires large labeled datasets; struggles with evolving patterns (e.g., new fraud types) |
| Unsupervised Learning [17] [18] [19] [20] [21] [22] [23] [24] [25] | Customer segmentation, anomaly detection, trend discovery | No labels required; adapts to dynamic data distributions | Hard to evaluate results; low interpretability (e.g., centroids in k-means) |
| Reinforcement Learning [26] [27] [28] [29] [30] [31] [32] [33] [34] [35] | Dynamic pricing, real-time recommendations, logistics optimization | Learns from interaction; handles sequential decisions in uncertain settings | Training overhead; complex reward design (e.g., RL + LSTM in chatbots) |
| Hybrid Learning [36] [37] [38] [39] [40] [41] [42] [43] [44] [45] [46] | Conversational AI, multi-source recommendation, hybrid fraud detection | Combines complementary strengths (e.g., RL + DL); robust to noisy inputs | High complexity; integration and tuning overhead (e.g., chatbots with memory units) |
| Meta-/Few-Shot Learning [47] [48] [49] [50] [51] [52] [53] [54] | Cold-start recommendation, adaptive ranking, new product personalization | Fast adaptation with few samples; cross-task generalization | Demanding training phase; real-time adaptation is resource-intensive |

gauge consumer sentiment toward pricing changes, ensuring price adjustments align with customer expectations and perceived product value [70], [71], [72].

### D. DEMAND FORECASTING AND INVENTORY MANAGEMENT

Accurate demand forecasting is integral to supply chain efficiency, minimizing stockouts and overstocking issues that lead to revenue losses. ML-based forecasting models surpass traditional statistical approaches by incorporating multidimensional data sources, including social media trends, macroeconomic indicators, and competitor inventory levels. Time-series forecasting algorithms, such as Prophet, LSTM networks, and attention-based transformers, predict demand fluctuations with high accuracy, enabling proactive inventory management [73], [74], [75].

The convergence of ML with Internet of Things (IoT) data has further revolutionized inventory optimization. Smart warehouses equipped with sensor-enabled systems transmit real-time inventory data to ML models, facilitating autonomous replenishment decisions. RL techniques optimize restocking strategies by considering procurement lead times, supplier reliability, and demand uncertainty. Furthermore, FL approaches are being explored to enhance cross-platform inventory synchronization, enabling collaborative inventory management among multiple retailers while preserving data privacy [76], [77], [78], [79].

### E. CUSTOMER SEGMENTATION AND BEHAVIORAL ANALYTICS

The ability to categorize customers based on their purchasing behaviours, preferences, and engagement levels is critical for targeted marketing and retention strategies. ML-powered clustering algorithms, including GMM and spectral clustering, identify latent customer segments that traditional demographic-based segmentation approaches fail to capture. By analysing high-dimensional user interaction data, ML models uncover hidden patterns that drive purchasing decisions, enabling hyper-personalized marketing campaigns [80], [81], [82].

Behavioural analytics extends beyond static segmentation by incorporating real-time interaction data to track evolving consumer preferences. RNNs and attention mechanisms capture sequential user behaviours, identifying shifts in shopping habits that signal churn risk or emerging product interests. These insights are instrumental in crafting automated engagement strategies, such as personalized email campaigns, tailored push notifications, and contextual advertising. The integration of sentiment analysis enhances behavioural analytics by extracting emotions from customer reviews, enabling brands to refine their messaging and product positioning accordingly [83], [84], [85], [86].

### F. CONVERSATIONAL AI AND AUTOMATED CUSTOMER SUPPORT

Adopting conversational AI has transformed customer support in e-commerce, providing intelligent, scalable solutions that enhance user experience. Traditional chatbots have evolved into advanced virtual assistants powered by transformer-based models, such as GPT (generative pretrained transformer) and BERT, which are capable of understanding natural language queries with contextual depth. These AI-driven assistants facilitate seamless interactions by resolving customer inquiries, assisting with product recommendations, and handling transactional processes autonomously [87], [88], [89].

Multimodal AI systems integrate text, voice, and visual recognition capabilities, enabling a more intuitive customer support experience. Visual search functionalities, powered by CNNs, allow users to find products by uploading images, providing continuity between digital and physical retail experiences. Sentiment-aware chatbots analyze user emotions in real-time, adapting their responses to provide empathetic and contextually relevant support. The integration of RL in chatbot development further enhances conversational flow by enabling adaptive dialogue management based on user interactions [90], [91], [92].

### G. REAL-TIME SEARCH OPTIMIZATION AND QUERY UNDERSTANDING

Search engines within e-commerce platforms rely on sophisticated ML models to enhance search accuracy and relevance. Traditional keyword-based retrieval systems have been replaced by semantic search models that understand user intent beyond literal keyword matches. DL architectures, including BERT-based retrieval models, enable contextual query understanding, improving search precision even in ambiguous queries [93], [94].

Personalized search ranking algorithms adapt search results based on individual user preferences, browsing history, and inferred intent. Multimodal search engines integrate textual, visual, and voice-based inputs to provide a seamless shopping experience across different interaction modes. RL in search ranking enables continuous optimization, refining result rankings based on user engagement metrics such as click-through rates and dwell time [95], [96], [97].

### H. SUPPLY CHAIN OPTIMIZATION AND LOGISTICS INTELLIGENCE

ML is pivotal in streamlining supply chain operations by optimizing logistics, transportation, and order fulfillment. Predictive analytics models anticipate shipment delays, enabling proactive mitigation strategies that enhance delivery efficiency. Route optimization algorithms powered by RL dynamically adjust delivery paths based on real-time traffic conditions and fuel efficiency metrics [98], [99], [100].

Integrating robotic process automation (RPA) with ML-driven decision-making has significantly improved warehouse automation. Autonomous robotic systems, guided by computer vision and Deep RL, optimize warehouse navigation and order-picking efficiency. Implementing digital twins in logistics enables virtual modelling of supply chain processes, allowing ML models to simulate and optimize operational workflows before real-world deployment [101], [102], [103].

Table 2 offers a comprehensive and comparative overview of how ML techniques are applied across major functional domains in e-commerce. Beyond listing methodologies and benefits, the table analytically maps each application area to its core functional scope, the ML techniques employed, the observed business benefits, and existing gaps or challenges. This side-by-side structure enables clearer insight into both the strengths and unresolved limitations of ML-driven e-commerce systems, guiding research and development directions.

## IV. CHALLENGES AND LIMITATIONS

Despite ML's transformative impact on e-commerce, several challenges and limitations hinder its seamless deployment and optimization. The increasing complexity of digital commerce, coupled with vast and evolving datasets, presents obstacles ranging from data privacy concerns and interpretability issues to computational constraints and scalability limitations. Addressing these challenges is crucial to ensuring robust, efficient, and ethical implementations of ML-driven e-commerce solutions.

### A. DATA PRIVACY, SECURITY, AND ETHICAL CONSIDERATIONS

One of the most pressing challenges in ML-driven e-commerce systems is the handling of sensitive user data while maintaining privacy, security, and ethical integrity. As businesses rely on vast amounts of personal information—ranging from transaction histories and browsing behaviours to location-based interactions—ensuring compliance with regulatory frameworks such as the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA) is imperative. Violations can lead to legal repercussions and loss of consumer trust, making privacy-preserving ML techniques, such as FL and differential privacy, essential for mitigating risks [104], [105], [106].

Furthermore, the ethical implications of AI-driven decision-making introduce concerns related to algorithmic bias, data exploitation, and consumer manipulation. Biased training datasets can lead to discriminatory pricing models, unfair targeting, and exclusionary recommendation systems that favour certain user demographics over others. Achieving transparency in ML models while preventing exploitative tactics such as dark patterns, which nudge users into unintended purchases, is an ongoing challenge in AI ethics within e-commerce [107], [108].

### B. MODEL INTERPRETABILITY AND TRUSTWORTHINESS

As e-commerce increasingly integrates DL-based systems, the lack of interpretability in these models raises concerns about decision transparency. Many ML models, particularly neural networks, lack transparency and interpretability, making it difficult to explain how specific recommendations, pricing decisions, or fraud detection alerts are generated. This opacity poses a barrier to regulatory compliance and user trust, particularly in domains such as financial transactions and personalized advertising [109], [110].

To enhance model interpretability, recent advances in Explainable AI (XAI) techniques, including SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations), attempt to provide post-hoc explanations for predictions. However, achieving both high accuracy and interpretability simultaneously remains a fundamental challenge. Businesses must balance predictive power with transparency to ensure accountability in automated decision-making [111], [112], [113], [114].

### C. COMPUTATIONAL COMPLEXITY AND SCALABILITY CONSTRAINTS

The exponential growth of e-commerce datasets necessitates highly efficient ML architectures capable of real-time processing. Traditional ML models struggle to scale when dealing with millions of users, billions of transactions, and

**TABLE 2.** Application domains and machine learning techniques in E-Commerce: comparative functional mapping.

| Application Domain | Scope & Functional Focus | ML Techniques | Benefits | Gaps / Open Issues |
|---|---|---|---|---|
| Personalized Recommendation and Consumer Behavior Modeling [55] [56] [57] [58] [59] [60] | Model user preferences to deliver personalized product suggestions in real time | Neural Collaborative Filtering, Transformers, Hybrid Models, RL | Increases engagement, conversion rates, and customer satisfaction; mitigates cold-start issues | Lacks explainability; domain adaptation is challenging across diverse product categories |
| Intelligent Fraud Detection and Risk Mitigation [61] [62] [63] [64] [65] [66] [67] | Identify anomalous or malicious behavior in transactions to prevent fraud | Anomaly Detection, GNNs, Semi-supervised Learning, Adversarial ML | Reduces fraudulent activity; increases trust and transaction safety | Real-time detection under imbalanced data remains difficult; evolving fraud tactics reduce accuracy |
| Dynamic Pricing and Revenue Optimization [68] [69] [70] [71] [72] | Automatically adjust prices based on market, user, and competitor signals | RL, Bayesian Optimization, Price Elasticity Modeling | Boosts revenue, improves price competitiveness, and personalizes pricing | High dependency on accurate demand signals; exploration-exploitation balance remains complex |
| Demand Forecasting and Inventory Management [73] [74] [75] [76] [77] [78] [79] | Predict future product demand to optimize inventory and supply chain flows | Time-Series Models (LSTM, Transformers), FL, RL | Minimizes stockouts and overstocking; enhances supply chain resilience | Accuracy affected by seasonal volatility and limited data in new product launches |
| Customer Segmentation and Behavioral Analytics [80] [81] [82] [83] [84] [85] [86] | Group customers based on behavioral patterns and characteristics | Clustering (GMM, Spectral), RNNs, Sentiment Analysis | Enables targeted promotions, loyalty strategies, and churn prediction | Defining cluster granularity is subjective; segment drift over time needs adaptation |
| Conversational AI and Automated Customer Support [87] [88] [89] [90] [91] [92] | Automate customer interaction with personalized, emotion-aware dialogue | Transformers (BERT, GPT), Sentiment-aware Chatbots, Multimodal AI | Enhances user satisfaction, reduces support costs, and shortens response time | Limited contextual memory; maintaining coherent long-term interactions remains unresolved |
| Real-Time Search Optimization and Query Understanding [93] [94] [95] [96] [97] | Personalize product search by understanding user intent and semantics | BERT-based Retrieval, Deep Ranking, RL-driven Search Optimization | Increases relevance and click-through rates; boosts product discovery | Contextual ambiguity in short queries; balancing latency vs. personalization quality |
| Supply Chain Optimization and Logistics Intelligence [98] [99] [100] [101] [102] [103] | Optimize logistics, routing, and warehouse management using predictive intelligence | Predictive Analytics, RL, Digital Twins, Computer Vision | Reduces delivery times and costs; improves warehouse throughput | Real-time coordination across heterogeneous logistics systems is still maturing |

continuously evolving product catalogues. DL approaches, while powerful, demand substantial computational resources. They often require high-performance graphics processing units (GPUs), tensor processing units (TPUs), or distributed cloud infrastructure to manage inference workloads effectively [115], [116], [117].

In recommendation systems, for example, real-time personalization demands low-latency model updates to reflect recent user behaviour. However, retraining large-scale models frequently is computationally prohibitive. Approximation techniques, such as knowledge distillation, model pruning, and quantization, are being explored to reduce complexity without significant accuracy loss. In addition, edge AI is emerging as a potential solution by shifting certain computations from cloud-based systems to on-device processing, reducing latency and enhancing efficiency in real-time e-commerce applications [118], [119], [120].

### D. DATA QUALITY, BIAS, AND GENERALIZATION ISSUES

ML models are only as good as the data they are trained on, making data quality a fundamental challenge in e-commerce. Noisy, inconsistent, or biased datasets can lead to erroneous predictions, negatively affecting recommendation accuracy, demand forecasting precision, and fraud detection robustness. A particular challenge arises from user behaviour drift, where consumer preferences evolve over time, rendering static models ineffective [121], [122].

Ensuring that models generalize well across diverse customer segments and geographical regions is also a significant hurdle. An ML model trained on data from one market may fail to perform optimally in another due to cultural, economic, or seasonal variations. Transfer learning techniques, domain adaptation strategies, and continuous online learning approaches are increasingly being explored to enable models to adjust dynamically to shifting patterns. However, these techniques introduce additional computational complexity and the risk of catastrophic forgetting—a phenomenon where a model abruptly loses performance on previously learned tasks while adapting to new information [123], [124], [125], [126].

### E. FRAUD ADAPTATION AND ADVERSARIAL ATTACKS

Fraud detection systems must constantly evolve to counteract the increasingly sophisticated tactics employed by cybercriminals. Traditional fraud detection models often rely on static heuristics, but as fraudsters develop new attack vectors, such models become ineffective. The adversarial nature of fraud necessitates ML systems that are continuously updated to detect emerging threats [127], [128].

A major challenge in fraud detection is the adversarial nature of the problem—attackers deliberately exploit weaknesses in ML models, launching adversarial attacks that manipulate transactional data or deceive fraud classification algorithms. Techniques such as adversarial retraining, robust feature engineering, and ensemble-based anomaly detection are being deployed to counteract these threats. However, fraudsters frequently adapt, requiring constant innovation in defence strategies [129], [130], [131].

Moreover, fraud detection models face the dilemma of minimizing false positives, where legitimate transactions are incorrectly flagged as fraud, while still maintaining a high true positive rate. Excessive false positives can lead to revenue loss due to rejected transactions, while false negatives allow fraudulent activities to persist undetected. Finding the optimal trade-off remains an ongoing challenge in e-commerce security [132], [133].

### F. REAL-TIME PROCESSING AND LATENCY CONSTRAINTS

E-commerce platforms operate in a high-speed digital environment where real-time decision-making is essential. From dynamically adjusting product recommendations to processing fraud detection alerts, ML models must execute complex computations in milliseconds to prevent delays that impact user experience. Achieving this level of responsiveness is particularly challenging for DL models, which require multiple layers of computation to generate predictions [134], [135], [136].

Deploying ML solutions that meet real-time performance requirements involves a combination of low-latency inference architectures, optimized data pipelines, and caching mechanisms. Graph-based recommendation systems, for instance, require efficient traversal operations to serve personalized recommendations instantly. Content-based filtering algorithms must rapidly adapt to newly introduced products, ensuring fresh and relevant suggestions for users. However, balancing real-time processing speed with predictive accuracy is an ongoing challenge, particularly when scaling ML solutions to global e-commerce ecosystems [137], [138], [139], [140].

### G. INTEGRATION COMPLEXITY AND MAINTENANCE CHALLENGES

Adopting ML in e-commerce requires seamless integration into existing business workflows, which can be complex due to compatibility issues between legacy systems and modern AI architectures. Many traditional e-commerce platforms are built on monolithic architectures that do not easily accommodate real-time ML models, necessitating expensive infrastructure upgrades or complete system overhauls [141], [142].

Additionally, ML models require continuous monitoring, retraining, and fine-tuning to maintain optimal performance. Model drift, where a model's predictive power deteriorates over time due to shifts in user behaviour or market trends, necessitates proactive retraining strategies. Implementing automated MLOps (ML Operations) frameworks is crucial for managing model lifecycle processes, including data preprocessing, feature engineering, hyperparameter tuning, and version control. However, implementing MLOps comes with significant resource requirements, particularly for smaller e-commerce businesses that may lack dedicated AI engineering teams [143], [144], [145], [146].

Table 3 provides a structured analysis of the key obstacles hindering the effective deployment of ML in e-commerce. It categorizes major challenges, highlights their direct impact on e-commerce systems, notes limitations, and outlines potential solutions that can mitigate these issues.

## V. FUTURE RESEARCH DIRECTIONS

As ML continues to transform e-commerce, emerging technologies are opening new frontiers that extend beyond predictive analytics toward generative, adaptive, and decentralized intelligence. Rapid advancements in generative AI, FL, neurosymbolic systems, and quantum-enhanced optimization offer novel tools to address longstanding challenges in scalability, personalization, interpretability, and trust. This section outlines key research directions that promise to reshape the technical and operational foundations of next-generation digital commerce.

### A. FEDERATED AND PRIVACY-PRESERVING LEARNING

The growing concerns surrounding data privacy, regulatory compliance, and ethical AI necessitate research into FL and privacy-enhancing ML techniques. Conventional centralized ML models require aggregating vast amounts of user data on a central server, raising concerns about security breaches, unauthorized data exploitation, and compliance with stringent privacy laws such as the GDPR. FL, which enables decentralized model training across multiple edge devices without exposing raw data, presents a promising solution. By allowing models to learn from distributed data while preserving privacy, this approach mitigates security risks and fosters user trust [147], [148].

However, FL introduces challenges related to communication overhead, heterogeneity in user data distributions, and model aggregation techniques. Research is needed to develop adaptive FL frameworks that optimize model synchronization across distributed nodes, ensuring efficiency and robustness. Additionally, integrating differential privacy and secure multi-party computation into FL architectures will further enhance security while maintaining predictive accuracy. Future advancements in privacy-preserving ML will drive the adoption of user-centric AI, enabling e-commerce platforms to leverage personalized insights while adhering to ethical AI standards [149], [150], [151].

### B. NEUROSYMBOLIC AI FOR EXPLAINABLE AND GENERALIZABLE LEARNING

Despite significant progress in DL, many ML models deployed in e-commerce remain opaque, making interpretation and reasoning difficult. Neurosymbolic AI, which combines the strengths of deep neural networks with symbolic reasoning, offers a compelling research direction to enhance model transparency and generalization. Unlike purely data-driven ML systems that rely on pattern recognition, neurosymbolic architectures integrate logical inference mechanisms, enabling models to reason about cause-and-effect relationships within consumer interactions [152], [153], [154].

For instance, in dynamic pricing optimization, a neurosymbolic AI system could incorporate economic theories and behavioural patterns alongside statistical learning, leading to more explainable and context-aware price adjustments. Similarly, in fraud detection, integrating symbolic rules with anomaly detection models can improve fraud identification by ensuring that predictions align with established risk factors rather than relying solely on statistical correlations.

**TABLE 3. Comparison of challenges and limitations in machine learning for E-Commerce.**

| Challenge | Impact on E-Commerce Systems | Potential Solutions | Limitations |
|---|---|---|---|
| Data Privacy, Security, and Ethical Considerations [104] [105] [106] [107] [108] | Compliance with regulations such as GDPR and CCPA requires strict data handling. Risk of algorithmic bias leads to unfair outcomes. Consumer trust is impacted by privacy violations and exploitative AI strategies. | FL and differential privacy to enhance security; AI fairness techniques to reduce bias; transparent AI frameworks to improve accountability. | FL increases computational and communication costs. AI fairness methods require careful dataset balancing, which may not always be feasible. Privacy measures can reduce model performance. |
| Model Interpretability and Trustworthiness [109] [110] [111] [112] [113] [114] | DL models often operate as black boxes, making it difficult to explain decisions in fraud detection, pricing, and recommendations. Low transparency reduces regulatory compliance and user trust. | XAI methods, such as SHAP and LIME; hybrid models combining interpretable rule-based approaches with DL; model audits to ensure fairness and reliability. | XAI techniques provide post-hoc explanations but do not inherently improve model transparency. Hybrid models can be computationally expensive. Interpretability may reduce model complexity, leading to lower accuracy. |
| Computational Complexity and Scalability Constraints [115] [116] [117] [118] [119] [120] | Large-scale ML models require high-performance computing, making real-time personalization and demand forecasting resource-intensive. Cloud costs and latency issues impact system efficiency. | Model compression techniques (pruning, quantization); edge AI for on-device processing; distributed ML frameworks to optimize resource usage. | Model compression can reduce accuracy. Edge AI requires specialized hardware. Distributed ML introduces synchronization and coordination challenges in large-scale deployments. |
| Data Quality, Bias, and Generalization Issues [121] [122] [123] [124] [125] [126] | Noisy and imbalanced datasets reduce model accuracy. Shifting user behaviours leads to model drift, requiring frequent retraining. Poor generalization across different markets affects performance. | Active learning and transfer learning to adapt models dynamically; data augmentation techniques for balanced training; CL frameworks to prevent catastrophic forgetting. | Active learning requires labelled data acquisition, which is costly. Transfer learning may not generalize well across domains. CL risks catastrophic forgetting if improperly tuned. |
| Fraud Adaptation and Adversarial Attacks [127] [128] [129] [130] [131] [132] [133] | Fraudsters continuously evolve tactics, reducing the effectiveness of static fraud detection models. Adversarial attacks manipulate ML models to evade detection. False positives in fraud detection lead to revenue loss. | Adversarial training and robust anomaly detection techniques; ensemble learning to improve fraud classification; continuous model updates and feedback loops. | Adversarial training increases model complexity and training time. Ensemble learning may lead to high computational overhead. Continual model updates can introduce operational delays and high resource usage. |
| Real-Time Processing and Latency Constraints [134] [135] [136] [137] [138] [139] [140] | ML-powered recommendations, search ranking, and fraud detection must operate within milliseconds to prevent delays in user experience. Complex DL architectures slow down inference. | Low-latency AI architectures using optimized caching and indexing; precomputed embeddings for rapid inference; graph-based models for efficient retrieval. | Low-latency optimization can compromise accuracy. Precomputed embeddings require frequent updates to reflect changing data. Graph-based models may suffer from scalability issues in large datasets. |
| Integration Complexity and Maintenance [141] [142] [143] [144] [145] [146] | Traditional e-commerce infrastructures may not support real-time ML model deployment. Continuous model monitoring and retraining increase operational complexity. | MLOps pipelines for automation of model updates; scalable cloud-based ML deployment; microservices architecture to integrate ML components seamlessly. | MLOps implementation requires specialized engineering skills. Cloud-based deployments introduce vendor lock-in risks. Microservices increase system complexity and potential latency overhead. |

Research into neurosymbolic AI will drive the development of interpretable, reasoning-driven ML models that bridge the gap between empirical learning and structured knowledge representation [155], [156], [157].

### C. QUANTUM MACHINE LEARNING FOR LARGE-SCALE OPTIMIZATION

As e-commerce datasets continue to grow in size and complexity, traditional ML algorithms face computational bottlenecks in handling large-scale optimization problems. Quantum computing, which leverages the principles of superposition and entanglement to perform parallel computations, presents a transformative opportunity for ML-driven e-commerce applications. QML has the potential to accelerate model training, enhance combinatorial optimization tasks, and solve high-dimensional problems with unprecedented efficiency [158], [159].

In the context of recommendation systems, QML could revolutionize personalized search and ranking algorithms by performing faster similarity computations across vast product catalogues. For inventory and logistics optimization, quantum-enhanced RL could significantly improve supply chain management, reducing inefficiencies in routing, demand forecasting, and warehouse allocation. However, the practical implementation of QML remains in its infancy, necessitating extensive research into hybrid quantum-classical algorithms that bridge quantum computing with existing ML infrastructures. Developing quantum-ready ML architectures that integrate with current e-commerce platforms will be a key research priority as quantum hardware matures [160], [161], [162], [163].

### D. MULTIMODAL AI FOR ENHANCED USER INTERACTION

The modern e-commerce experience is no longer limited to text-based interactions but incorporates images, voice commands, and real-time engagement across multiple modalities. Traditional recommendation systems rely primarily on transactional and behavioural data, but the integration of multimodal AI, which fuses textual, visual, and auditory data, offers a new frontier for personalized user experiences [164], [165].

Future research will focus on developing transformer-based multimodal architectures capable of processing and aligning diverse data sources. For instance, a shopper browsing an online store could receive AI-driven recommendations not only based on previous purchases but also on visual preferences, voice queries, and facial expressions. The convergence of NLP, computer vision, and affective computing will enable deeper user engagement, allowing ML models to infer user intent more holistically. Moreover, real-time emotion-aware AI will enable adaptive personalization, where AI systems dynamically adjust recommendations based on user sentiment and micro-expressions during an interaction [166], [167], [168].

Advancements in multimodal fusion techniques will be critical in enabling AI systems to interpret and synthesize complex user interactions seamlessly. Future research must address challenges such as cross-modal alignment, efficient representation learning, and real-time inference latency to unlock the full potential of multimodal AI in e-commerce [169], [170].

### E. SELF-SUPERVISED AND CONTINUAL LEARNING FOR ADAPTIVE AI

The fast-evolving nature of consumer behaviour and product trends presents a fundamental challenge for traditional ML models, which require periodic retraining on newly labelled datasets. SSL (self-supervised learning) and CL present promising research directions to enable AI systems that learn continuously from evolving data streams without requiring extensive manual annotations [171], [172].

SSL enables models to extract meaningful representations from unlabeled data, reducing dependency on large annotated datasets while improving generalization across domains. For instance, SSL-based recommendation systems can autonomously learn user preferences by identifying patterns in implicit interactions, such as scrolling behaviour, session duration, and cursor movement. This eliminates the need for extensive labelled feedback while enabling real-time personalization [173], [174], [175].

Conversely, continuous learning equips AI systems with the ability to retain previously learned knowledge while incorporating new information. Unlike traditional ML models that suffer from catastrophic forgetting, CL frameworks ensure that AI-driven e-commerce platforms adapt to new market trends, emerging product categories, and evolving fraud tactics without losing prior knowledge. Research efforts in memory-augmented neural networks and dynamic weight adaptation will be crucial in developing AI systems that learn persistently, maintaining relevance over time without requiring complete retraining [176], [177], [178], [179].

### F. DECENTRALIZED AND BLOCKCHAIN-BASED AI FOR TRUST AND SECURITY

The reliance on centralized AI systems in e-commerce raises concerns about trust, transparency, and data monopolization, prompting research into decentralized ML frameworks powered by blockchain technology. Blockchain-based AI has the potential to enhance security in fraud detection, ensure transparent AI decision-making, and establish decentralized marketplaces where consumers retain control over their data [180], [181].

Future research in this area will explore the integration of smart contracts with ML-based fraud detection, where AI-driven risk assessments are recorded on an immutable ledger, preventing tampering and ensuring auditability. Additionally, decentralized autonomous recommendation systems (DARS) could allow users to personalize their shopping experiences without relying on centralized platforms that monetize user data [182], [183], [184].

Addressing the computational inefficiencies of blockchain-based AI remains a critical research challenge. Key areas of exploration will include optimizing consensus mechanisms, reducing the energy consumption of decentralized AI networks, and developing scalable federated blockchain architectures. As e-commerce platforms seek to enhance user trust and reduce reliance on intermediaries, decentralized AI will play a pivotal role in shaping the next generation of secure and user-centric digital marketplaces [185], [186], [187].

### G. GENERATIVE AI FOR INTELLIGENT E-COMMERCE

Recent advances in generative AI and large foundation models, such as GPT, Claude, and PaLM (Pathways Language Model), are poised to redefine the operational and experiential landscape of e-commerce. These models exhibit powerful capabilities in natural language generation, image synthesis, multimodal understanding, and contextual dialogue, enabling a new class of intelligent services. Applications include dynamic product description generation, AI-generated visual content, personalized marketing narratives, and conversational agents that adapt to individual customer intents with high semantic fidelity. These capabilities introduce a paradigm shift from static personalization toward generative personalization at scale [188], [189].

Despite these promising use cases, integrating generative models into e-commerce ecosystems raises open challenges in controllability, latency, fine-tuning efficiency, and alignment with brand and legal constraints. Research is needed to develop domain-specific foundation models that are optimized for product catalogs, customer reviews, behavioral signals, and regulatory frameworks. Equally important is the investigation of techniques for hallucination mitigation, brand-safe generation, and low-resource deployment, particularly for mobile and edge-based commerce scenarios where inference latency is critical. The trade-off between generation quality, real-time performance, and interpretability must also be addressed through novel architecture and compression strategies [190], [191].

Future directions include generative agents that support multi-turn interactions, negotiation, and adaptive product

**TABLE 4. Future research directions in machine learning for E-Commerce.**

| Research Area | Key Challenges | Potential Solutions | Expected Impact on E-Commerce |
|---|---|---|---|
| Federated and Privacy-Preserving Learning [147] [148] [149] [150] [151] | Communication overhead, data heterogeneity, model aggregation inefficiencies | Adaptive FL, differential privacy, secure multi-party computation | Ensures data security, enhances compliance with privacy regulations, enables decentralized AI-driven personalization |
| Neurosymbolic AI for Explainable and Generalizable Learning [152] [153] [154] [155] [156] [157] | Black-box nature of DL, lack of reasoning-based decision-making | Hybrid DL with symbolic logic, rule-based inference integration | Improves interpretability of AI decisions, enables context-aware dynamic pricing and fraud detection |
| QRL for Large-Scale Optimization [158] [159] [160] [161] [162] [163] | High computational costs of traditional ML, complexity in large-scale data optimization | Hybrid quantum-classical algorithms, quantum-enhanced RL | Accelerates training times, optimizes logistics, enables scalable real-time recommendations |
| Multimodal AI for Enhanced User Interaction [164] [165] [166] [167] [168] [169] [170] | Cross-modal data alignment, high computational demands for real-time inference | Transformer-based multimodal architectures, multimodal fusion networks | Enhances personalized shopping experiences, improves recommendation accuracy using vision, text, and speech data |
| SSL and CL for Adaptive AI [171] [172] [173] [174] [175] [176] [177] [178] [179] | Model drift, high dependence on labelled data, catastrophic forgetting in long-term learning | Memory-augmented networks, adaptive learning rate tuning, contrastive SSL | Reduces the need for frequent retraining, enables AI to continuously learn from user behaviour |
| Decentralized and Blockchain-Based AI for Trust and Security [180] [181] [182] [183] [184] [185] [186] [187] | High computational overhead of blockchain, lack of AI integration in decentralized systems | Federated blockchain models, smart contracts for AI transparency | Enhances security in fraud detection, establishes decentralized recommendation systems, ensures trust in AI-driven transactions |
| Generative AI for Intelligent E-Commerce [188] [189] [190] [191] [192] [193] [194] | Model hallucination, lack of controllability, latency, domain adaptation challenges | Domain-specific fine-tuning, hallucination mitigation, hybrid architectures, lightweight inference for edge deployment | Enables dynamic content generation, personalized dialogue, synthetic data creation, and conversational product discovery with brand-aligned, real-time interaction |

discovery by combining generative transformers with RL and structured knowledge bases. Additionally, generative models can be leveraged to synthesize rare user behaviors and simulate fraud patterns, contributing to more robust training pipelines. As generative AI continues to evolve, its convergence with CL, neurosymbolic systems, and decentralized inference will likely result in deeply personalized, trustworthy, and context-aware e-commerce platforms [192], [193], [194].

Table 4 provides a research-focused overview of emerging ML advancements that will shape the future of digital commerce. It categorizes various key research areas, identifies the primary challenges associated with each, proposes potential solutions, and outlines the expected impact on e-commerce applications.

## VI. DISCUSSION

This survey reveals a transition in how ML is adopted in e-commerce, moving from isolated use cases to integrated systems that prioritize adaptability, real-time inference, and privacy. Advances such as FL, neurosymbolic models, CL, and transformer-based generative architectures are reshaping tasks like personalization, fraud detection, and customer segmentation by enabling more context-aware and resilient decision-making.

Table 5 synthesizes prior surveys. The authors in [195] and [198] review core ML and DL models for standard applications, yet they overlook architectural issues such as privacy and sustainability. Besides, [196] focuses narrowly on blockchain-ML integration, while [197] and [200] contribute insights into churn and retention but offer limited architectural generalizability. Also, [201] aligns ML models with business goals but lacks deployment-oriented analysis. Reference [199] emphasizes analytics pipelines but downplays interpretability and robustness. Moreover, [202] examines security and privacy trends, yet their analysis is not commerce-specific. Finally, [203] discusses e-commerce evolution but does not engage with model-level ML trade-offs.

The present survey addresses those limitations by linking learning paradigms to both commercial functionality and system-level constraints, such as latency, explainability, and regulatory compliance. It also incorporates underrepresented methods, such as neurosymbolic inference and QRL, which are largely absent from commerce-focused reviews. Rather than emphasizing algorithmic classification alone, the analysis considers practical trade-offs in real-world ML deployment.

In addition to paradigm-based categorization, lifecycle-aware modeling provides a valuable perspective. As reviewed in [204], aligning ML systems to specific stages in the customer journey (e.g., onboarding, retention, reactivation) enhances targeting precision and model relevance. Future surveys might benefit from hybrid frameworks that map learning methods not only to technical constraints but also to user behavior states.

Emerging ML techniques are not without operational challenges. FL introduces synchronization overhead and hardware distribution constraints. QML remains experimental, with issues in reproducibility and accessibility. As highlighted by [205], the environmental impact of large-scale training and frequent model updates—especially

**TABLE 5.** Summary and thematic scope of reviewed survey articles.

| Reference | Description |
|---|---|
| [195] | Offers a comprehensive survey of ML and DL techniques applied to e-commerce between 2018 and 2023. It covers supervised and unsupervised ML models, CNNs, RNNs, GANs, transformers, and BERT, and maps them to applications such as sentiment analysis, recommendation, fraud detection, and image recognition. It also introduces a detailed taxonomy and discusses challenges like overfitting, interpretability, multimodality, and personalization. |
| [196] | Explores the integration of blockchain and ML in e-commerce and other industries, discussing benefits, challenges, and future research directions while evaluating the impact on security, privacy, and decision-making. |
| [197] | Focuses on customer churn prediction in e-commerce using ML and data mining techniques. It highlights predictive models, performance metrics, and methodologies for improving customer retention strategies. |
| [198] | Provides a comprehensive review of ML developments in e-commerce, analyzing applications in customer behaviour analysis, inventory management, and fraud detection, discussing challenges and future directions. |
| [199] | Examines the role of data analytics and ML in e-commerce, emphasizing their impact on automation, decision-making, and personalization. It also explores existing challenges, such as data quality, algorithm suitability, and implementation costs. |
| [200] | Investigates ML frameworks for customer retention and propensity modelling in e-commerce, presenting advanced predictive models and analyzing their effectiveness in improving customer engagement and lifetime value. |
| [201] | Presents an up-to-date systematic literature review of ML applications in e-commerce, analyzing 70 articles over a five-year period. It introduces a novel taxonomy linking ML techniques to key business goals such as recommendation systems, fraud detection, customer retention, and purchase prediction, and discusses the impact of ML on profit, personalization, and operational efficiency. |
| [202] | Conducts a large-scale systematic review of over 9,000 publications on AI in cybersecurity and privacy using the PRISMA framework and BERTopic modeling. It identifies 14 core AI-driven themes, including intrusion detection, FL for privacy, adversarial ML, and blockchain-based security, providing a roadmap of challenges and opportunities for AI research in secure digital systems. |
| [203] | Gives a conceptual overview of the evolution of e-commerce, highlighting how ML and AI have transformed digital commerce processes. It covers use cases in supply chain management, customer relationship management, and revenue models. The chapter blends theoretical discussion with applied insights and real-world examples from the retail sector. |

in FL settings—raises important sustainability concerns that are underexplored in current literature.

This survey offers guidance for model selection under deployment constraints for practitioners. For example, FL can enable privacy-preserving personalization in compliance-heavy domains, while lightweight transformers may support mobile-first fraud detection. Trade-offs between interpretability, accuracy, and latency should inform system design in live commerce environments.

Theoretically, the paper highlights directions for deeper exploration, including hybrid symbolic–subsymbolic architectures for personalization, and CL systems that adapt to changing consumer behavior without repeated retraining. It also points to the need for cross-disciplinary integration across AI, behavioral science, and digital policy to shape responsible e-commerce innovation.

Future commerce platforms will likely intersect with technologies such as immersive interfaces, decentralized identity, and real-time generative content. These shifts demand ML systems that are not only adaptive but also secure and explainable by design. As paper [202] emphasizes, trust must be engineered into the ML lifecycle, from deployment training, rather than appended after the fact.

While this taxonomy introduces a structured and functional lens for mapping ML technologies in e-commerce, it is not exhaustive. As the field evolves, future work should incorporate ethical, social, and environmental perspectives alongside computational considerations. Flexible, multi-stakeholder taxonomies will be essential for guiding responsible AI development in commercial ecosystems.

In summary, this discussion offers a comparative lens on existing literature, distills actionable insights for both implementation and research, and highlights how this survey complements prior work with a constraint-aware, multi-paradigm taxonomy. Future extensions should aim to integrate environmental metrics, stakeholder roles, and adaptive learning stages into commerce-specific AI systems.

## VII. CONCLUSION

ML has revolutionized e-commerce by enabling intelligent decision-making, enhancing user experience, and optimizing business operations. This survey has provided a structured exploration of ML applications in e-commerce, methodologies that drive these innovations, inherent challenges, and future research directions that promise to redefine digital commerce. The increasing complexity of online markets necessitates the integration of adaptive, scalable, and XAI models that cater to dynamic consumer behaviours, security threats, and computational constraints.

While supervised and unsupervised learning remain fundamental to predictive analytics and pattern discovery, RL has emerged as a powerful tool for optimizing pricing strategies, recommendation systems, and logistics management. Hybrid and neurosymbolic AI models are bridging the gap between interpretability and predictive accuracy, addressing long-standing concerns regarding algorithmic transparency. Additionally, advancements in FL and privacy-preserving AI techniques are mitigating risks associated with data security and compliance, allowing e-commerce platforms to leverage vast consumer datasets while adhering to regulatory frameworks.

Despite these advancements, significant obstacles persist, ranging from real-time inference latency and model scalability to ethical considerations surrounding AI-driven decision-making. Fraud detection systems must continuously evolve to counteract adversarial attacks, while recommendation engines require continuous learning mechanisms to adapt to shifting consumer preferences. Integrating SSL and CL is a promising avenue for enhancing model robustness without excessive retraining. However, these techniques introduce challenges related to computational efficiency and long-term knowledge retention, necessitating further research into optimization strategies that balance adaptability with operational feasibility.

Looking ahead, the convergence of QML, blockchain-based AI, and multimodal intelligence will reshape the landscape of e-commerce. Quantum-enhanced models hold the potential to accelerate complex optimizations in personalization and logistics, while decentralized AI frameworks will redefine data ownership, transparency, and fraud mitigation strategies. Multimodal AI, integrating textual, visual, and auditory data, will drive the next wave of immersive and hyper-personalized shopping experiences. These emerging technologies will enhance predictive accuracy and improve the overall efficiency and security of e-commerce ecosystems.

To fully harness the potential of ML in e-commerce, interdisciplinary collaboration between AI researchers, behavioural economists, cybersecurity experts, and industry practitioners is imperative. The adoption of ML should be guided by ethical principles that prioritize fairness, inclusivity, and user-centric design. Future innovations must address existing limitations while fostering scalable, explainable, and privacy-conscious AI models that align with the evolving needs of digital commerce. By advancing research in these critical areas, ML will continue to drive the transformation of e-commerce, shaping a more intelligent, secure, and user-driven marketplace in the years to come.

## REFERENCES

[1] M. Iqbal, "Machine learning applications in e-commerce," *Org., Bus. Manage.*, vol. 65, pp. 1–15, Jan. 2022.
[2] I. Oktaviani, E. Purawanto, and T. Triana, "Analysis of AI-based big data for strategic decision-making in e-commerce," in *Proc. Int. Conf. Sci. Health Technol.*, Sep. 2024, vol. 5, no. 1, p. 4192.
[3] R. El Youbi, F. Messaoudi, and M. Loukili, "Machine learning-driven dynamic pricing strategies in e-commerce," in *Proc. 14th Int. Conf. Inf. Commun. Syst. (ICICS)*, Nov. 2023, pp. 1–5.
[4] R. Khurana, "Fraud detection in eCommerce payment systems: The role of predictive AI in real-time transaction security and risk management," *Int. J. Appl. Mach. Learn. Comput. Intell.*, vol. 10, no. 6, pp. 1–32, Jun. 2020.
[5] J. Hu, R. Hu, Z. Wang, D. Li, J. Wu, L. Ren, Y. Zang, Z. Huang, and M. Wang, "Collaborative fraud detection: How collaboration impacts fraud detection," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 8891–8899.
[6] T. Karunaratne, "Machine learning and big data approaches to enhancing e-commerce anomaly detection and proactive defense strategies in cybersecurity," *J. Adv. Cybersecurity Sci., Threat Intell., Countermeasures*, vol. 7, no. 12, pp. 1–16, 2023.
[7] V. M. Reddy and L. N. Nalla, "Real-time data processing in e-commerce: Challenges and solutions," *Int. J. Adv. Eng. Technol. Innov.*, vol. 1, no. 3, pp. 297–325, 2024.
[8] R. Raman, R. Kowalski, K. Achuthan, A. Iyer, and P. Nedungadi, "Navigating artificial general intelligence development: Societal, technological, ethical, and brain-inspired pathways," *Sci. Rep.*, vol. 15, no. 1, pp. 1–22, Mar. 2025.
[9] R. Jhangiani, D. Bein, and A. Verma, "Machine learning pipeline for fraud detection and prevention in e-commerce transactions," in *Proc. IEEE 10th Annu. Ubiquitous Comput., Electron. Mobile Commun. Conf. (UEMCON)*, Oct. 2019, pp. 0135–0140.
[10] A. Panarese, G. Settanni, V. Vitti, and A. Galiano, "Developing and preliminary testing of a machine learning-based platform for sales forecasting using a gradient boosting approach," *Appl. Sci.*, vol. 12, no. 21, p. 11054, Oct. 2022.
[11] S. Wang, C. Liu, X. Gao, H. Qu, and W. Xu, "Session-based fraud detection in online e-commerce transactions using recurrent neural networks," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discovery Databases*, Skopje, Macedonia. Cham, Switzerland: Springer, Jan. 2017, pp. 241–252.
[12] D. Wang, J. Lin, P. Cui, Q. Jia, Z. Wang, Y. Fang, Q. Yu, J. Zhou, S. Yang, and Y. Qi, "A semi-supervised graph attentive network for financial fraud detection," in *Proc. IEEE Int. Conf. Data Mining (ICDM)*, Nov. 2019, pp. 598–607.
[13] T. Hu, Q. Guo, X. Shen, H. Sun, R. Wu, and H. Xi, "Utilizing unlabeled data to detect electricity fraud in AMI: A semisupervised deep learning approach," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 11, pp. 3287–3299, Nov. 2019.
[14] K. N. Prasanthi, R. E. Madhavi, D. N. S. Sabarinadh, and B. Sravani, "A novel approach for sentiment analysis on social media using BERT & ROBERTA transformer-based models," in *Proc. IEEE 8th Int. Conf. Converg. Technol. (I2CT)*, Apr. 2023, pp. 1–6.
[15] A. Sharma, N. Patel, and R. Gupta, "Leveraging LSTM and prophet models for enhanced AI-driven demand prediction in e-commerce," *Eur. Adv. Artif. Intell. J.*, vol. 3, no. 2, pp. 45–58, Dec. 2021.
[16] H. Xu and Y. Lv, "Mining and application of tourism online review text based on natural language processing and text classification technology," *Wireless Commun. Mobile Comput.*, vol. 2022, pp. 1–13, May 2022.
[17] B. Shen, "E-commerce customer segmentation via unsupervised machine learning," in *Proc. 2nd Int. Conf. Comput. Data Sci.*, Jan. 2021, pp. 1–7.
[18] S. Chandra, S. Verma, W. M. Lim, S. Kumar, and N. Donthu, "Personalization in personalized marketing: Trends and ways forward," *Psychol. Marketing*, vol. 39, no. 8, pp. 1529–1562, Aug. 2022.
[19] N. D. Sugiharto, D. Elbert, J. Arnold, I. S. Edbert, and D. Suhartono, "Mall customer clustering using Gaussian mixture model, K-means, and BIRCH algorithm," in *Proc. 6th Int. Conf. Inf. Commun. Technol. (ICOIACT)*, Nov. 2023, pp. 212–217.
[20] F. Anowar, S. Sadaoui, and B. Selim, "Conceptual and empirical comparison of dimensionality reduction algorithms (PCA, KPCA, LDA, MDS, SVD, LLE, ISOMAP, LE, ICA, t-SNE)," *Comput. Sci. Rev.*, vol. 40, May 2021, Art. no. 100378.

[21] Y. Hu, "Research on e-commerce short video recommendation system based on user behavior data mining and feature extraction," in *Proc. Int. Conf. Comput., Inf. Process. Adv. Educ. (CIPAE)*, Aug. 2024, pp. 119–126.

[22] Ö. Özkum, "Credit card fraud detection with autoencoders, one-class svms and isolation forests," M.S. thesis, Graduate School Natural Appl. Sci., Middle East Tech. Univ., Ankara, Türkiye, 2023.

[23] Y. Ding, W. Kang, J. Feng, B. Peng, and A. Yang, "Credit card fraud detection based on improved variational autoencoder generative adversarial network," *IEEE Access*, vol. 11, pp. 83680–83691, 2023.

[24] J. Shan, *Synthetic Data Generation for Fraud Detection*. Los Angeles, CA, USA: University of California, 2023.

[25] A. Iqbal, E. Ahmed, A. Rahman, and M. R. H. Ontor, "Enhancing fraud detection and anomaly detection in retail banking using generative ai and machine learning models," *Amer. J. Eng. Technol.*, vol. 6, no. 11, pp. 78–91, Nov. 2024.

[26] N. Chopra, A. Patel, N. Singh, and V. Sharma, "Leveraging reinforcement learning and neural networks for optimized dynamic pricing strategies in e-commerce," *Int. J. AI Advancement*, vol. 9, no. 4, pp. 45–58, Dec. 2020.

[27] H. Meisheri, V. Baniwal, N. N. Sultana, B. Ravindran, and H. Khadilkar, "Reinforcement learning for multi-objective optimization of online decisions in high-dimensional systems," 2019, *arXiv:1910.00211*.

[28] J. Liu, Y. Zhang, X. Wang, Y. Deng, and X. Wu, "Dynamic pricing on e-commerce platform with deep reinforcement learning: A field experiment," 2019, *arXiv:1912.02572*.

[29] P. Famil Alamdar and A. Seifi, "A deep Q-learning approach to optimize ordering and dynamic pricing decisions in the presence of strategic customers," *Int. J. Prod. Econ.*, vol. 269, Mar. 2024, Art. no. 109154.

[30] X. Tang, Y. Chen, X. Li, J. Liu, and Z. Ying, "A reinforcement learning approach to personalized learning recommendation systems," *Brit. J. Math. Stat. Psychol.*, vol. 72, no. 1, pp. 108–135, Feb. 2019.

[31] N. Silva, H. Werneck, T. Silva, A. C. M. Pereira, and L. Rocha, "Multi-armed bandits in recommendation systems: A survey of the state-of-the-art and future directions," *Expert Syst. Appl.*, vol. 197, Jul. 2022, Art. no. 116669.

[32] M. Fu, L. Huang, A. Rao, A. A. Irissappane, J. Zhang, and H. Qu, "A deep reinforcement learning recommender system with multiple policies for recommendations," *IEEE Trans. Ind. Informat.*, vol. 19, no. 2, pp. 2049–2061, Feb. 2023.

[33] L. Kemmer, H. von Kleist, D. de Rochebouët, N. Tziortziotis, and J. Read, "Reinforcement learning for supply chain optimization," in *Proc. Eur. Workshop Reinforcement Learn.*, vol. 14, Oct. 2018, pp. 1–15.

[34] X. Liu, M. Hu, Y. Peng, and Y. Yang, "Multi-agent deep reinforcement learning for multi-echelon inventory management," *Prod. Oper. Manage.*, vol. 33, no. 12, Dec. 2022, Art. no. 10591478241305863.

[35] Y. Yan, A. H. F. Chow, C. P. Ho, Y.-H. Kuo, Q. Wu, and C. Ying, "Reinforcement learning for logistics and supply chain management: Methodologies, state of the art, and future opportunities," *Transp. Res. E, Logistics Transp. Rev.*, vol. 162, Jun. 2022, Art. no. 102712.

[36] R. Venkatesan and A. Sabari, "Deepsentimodels: A novel hybrid deep learning model for an effective analysis of ensembled sentiments in e-commerce and s-commerce platforms," *Cybern. Syst.*, vol. 54, no. 4, pp. 526–549, May 2023.

[37] D. D. Dasig, D. J. R. Calantoc, R. V. F. Guarin, M. A. B. Taduyo, C. N. Ferrer, E. E. Claricia, and G. E. Agus, "Predicting customer purchase decisions using data mining technique," in *Proc. IEEE 14th Int. Conf. Humanoid, Nanotechnol., Inf. Technol., Commun. Control, Environ., Manage. (HNICEM)*, Dec. 2022, pp. 1–6.

[38] Y. Afoudi, M. Lazaar, and M. A. Achhab, "Hybrid recommendation system combined content-based filtering and collaborative prediction using artificial neural network," *Simul. Model. Pract. Theory*, vol. 113, Dec. 2021, Art. no. 102375.

[39] E. F. Malik, K. W. Khaw, B. Belaton, W. P. Wong, and X. Chew, "Credit card fraud detection using a new hybrid machine learning architecture," *Mathematics*, vol. 10, no. 9, p. 1480, Apr. 2022.

[40] T. Zhou and H. Jiao, "Exploration of the stacking ensemble machine learning algorithm for cheating detection in large-scale assessment," *Educ. Psychol. Meas.*, vol. 83, no. 4, pp. 831–854, Aug. 2023.

[41] X. Niu, L. Wang, and X. Yang, "A comparison study of credit card fraud detection: Supervised versus unsupervised," 2019, *arXiv:1904.10604*.

[42] N. Esfandiari, K. Kiani, and R. Rastgoo, "Transformer-based generative chatbot using reinforcement learning," *J. AI Data Mining*, vol. 12, no. 3, pp. 349–358, Jul. 2024.

[43] K. R. Praneeth, T. S. Ruprah, J. N. Madhuri, A. L. Sreenivasulu, S. Shareefunnisa, and V. S. Rao, "Optimizing customer interactions: A BERT and reinforcement learning hybrid approach to chatbot development," *Int. J. Adv. Comput. Sci. Appl.*, vol. 15, no. 9, pp. 569–578, 2024.

[44] M. Ahmed, H. U. Khan, and E. U. Munir, "Conversational AI: An explication of few-shot learning problem in transformers-based chatbot systems," *IEEE Trans. Computat. Social Syst.*, vol. 11, no. 2, pp. 1888–1906, Apr. 2024.

[45] R. V. Joseph, A. Mohanty, S. Tyagi, S. Mishra, S. K. Satapathy, and S. N. Mohanty, "A hybrid deep learning framework with CNN and bi-directional LSTM for store item demand forecasting," *Comput. Electr. Eng.*, vol. 103, Oct. 2022, Art. no. 108358.

[46] A. K. Kalusivalingam, A. Sharma, N. Patel, and V. Singh, "Optimizing inventory management with AI: Leveraging deep reinforcement learning and neural networks for enhanced demand forecasting and stock replenishment," *Int. J. Artif. Intell. Mach. Learn.*, vol. 1, no. 1, p. 43, Dec. 2024.

[47] B. Wu, Z. Meng, Q. Zhang, and S. Liang, "Meta-learning helps personalized product search," in *Proc. ACM Web Conf.*, Apr. 2022, pp. 2277–2287.

[48] Y. Wang, Q. Yao, J. T. Kwok, and L. M. Ni, "Generalizing from a few examples: A survey on few-shot learning," *ACM Comput. Surveys*, vol. 53, no. 3, pp. 1–34, May 2021.

[49] A. Agnihotri and I. I. Raj, "Advanced deep reinforcement learning framework for dynamic pricing optimization in e-commerce marketplaces," in *Proc. 15th Int. Conf. Comput. Commun. Netw. Technol. (ICCCNT)*, Jun. 2024, pp. 1–6.

[50] M. Li, S. Hu, F. Zhu, and Q. Zhu, "Few-shot learning for cold-start recommendation," in *Proc. Joint Int. Conf. Comput. Linguistics, Lang. Resour. Eval. (LREC-COLING)*, May 2024, pp. 7185–7195.

[51] M. Kim, H. Song, Y. Shin, D. Park, K. Shin, and J.-G. Lee, "Meta-learning for online update of recommender systems," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2022, vol. 36, no. 4, pp. 4065–4074.

[52] D. Wang, M. Zhang, Y. Xu, W. Lu, J. Yang, and T. Zhang, "Metric-based meta-learning model for few-shot fault diagnosis under multiple limited data conditions," *Mech. Syst. Signal Process.*, vol. 155, Jun. 2021, Art. no. 107510.

[53] R. Yu, Y. Gong, X. He, Y. Zhu, Q. Liu, W. Ou, and B. An, "Personalized adaptive meta learning for cold-start user preference prediction," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, vol. 35, no. 12, pp. 10772–10780.

[54] Q. Wang, X. Liu, W. Liu, A.-A. Liu, W. Liu, and T. Mei, "MetaSearch: Incremental product search via deep meta-learning," *IEEE Trans. Image Process.*, vol. 29, pp. 7549–7564, 2020.

[55] F. Messaoudi and M. Loukili, "E-commerce personalized recommendations: A deep neural collaborative filtering approach," in *Operations Research Forum*, vol. 5. Cham, Switzerland: Springer, 2024, p. 5.

[56] Q. Sun, Y. Xue, and Z. Song, "Adaptive user interface generation through reinforcement learning: A data-driven approach to personalization and optimization," 2024, *arXiv:2412.16837*.

[57] T. Hagendorff, "Linking human and machine behavior: A new approach to evaluate training data quality for beneficial machine learning," *Minds Mach.*, vol. 31, no. 4, pp. 563–593, Dec. 2021.

[58] S. Raza and C. Ding, "Progress in context-aware recommender systems—An overview," *Comput. Sci. Rev.*, vol. 31, pp. 84–97, Feb. 2019.

[59] V. G. Morales-Murillo, D. Pinto, F. Perez-Tellez, and F. Rojas-Lopez, "A transformer-based multi-domain recommender system for e-commerce," *Int. J. Combinat. Optim. Problems Informat.*, vol. 15, no. 2, pp. 95–123, Jun. 2024.

[60] H. Li and D. Han, "A time-aware hybrid recommendation scheme combining content-based and collaborative filtering," *Frontiers Comput. Sci.*, vol. 15, no. 4, Aug. 2021, Art. no. 154613.

[61] J. Gupta, "Credit card fraud detection using machine learning algorithms," *Int. J. Sci. Res.*, vol. 12, no. 11, pp. 1774–1779, Nov. 2023.

[62] T. DeLise, "Deep semi-supervised anomaly detection for finding fraud in the futures market," 2023, *arXiv:2309.00088*.

[63] S. Carta, G. Fenu, D. Reforgiato Recupero, and R. Saia, "Fraud detection for e-commerce transactions by employing a prudential multiple consensus model," *J. Inf. Secur. Appl.*, vol. 46, pp. 13–22, Jun. 2019.

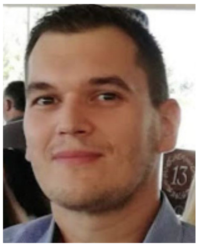
[64] A. Kotiyal, L. Hussein, A. Deepak, A. Rana, Manjunatha, K. K. Dixit, and R. A. Reddy, "Graph-based machine learning approaches for fraud detection in financial networks," in *Proc. 7th Int. Conf. Contemp. Comput. Informat. (IC3I)*, Sep. 2024, pp. 1714–1720.

[65] U. Fiore, A. De Santis, F. Perla, P. Zanetti, and F. Palmieri, "Using generative adversarial networks for improving classification effectiveness in credit card fraud detection," *Inf. Sci.*, vol. 479, pp. 448–455, Apr. 2019.
[66] M. Das, H. Luo, and J. C. Cheng, "Securing interim payments in construction projects through a blockchain-based framework," *Autom. Construct.*, vol. 118, Oct. 2020, Art. no. 103284.
[67] M. Thilagavathi, R. Saranyadevi, N. Vijayakumar, K. Selvi, L. Anitha, and K. Sudharson, "AI-driven fraud detection in financial transactions with graph neural networks and anomaly detection," in *Proc. Int. Conf. Sci. Technol. Eng. Manage. (ICSTEM)*, Apr. 2024, pp. 1–6.
[68] C. Yin and J. Han, "Dynamic pricing model of e-commerce platforms based on deep reinforcement learning," *Comput. Model. Eng. Sci.*, vol. 127, no. 1, pp. 291–307, 2021.
[69] X. Zhu, L. Jian, C. Xin, and Q. Zhao, "Deep Q-learning for multi-flight dynamic pricing: Maximizing revenue with a novel utility function in airline revenue management," *Comput. Ind. Eng.*, vol. 193, Jun. 2024, Art. no. 110302.
[70] P. Das, T. Pervin, B. Bhattacharjee, M. R. Karim, N. Sultana, M. S. Khan, M. A. Hosien, and F. Kamruzzaman, "Optimizing real-time dynamic pricing strategies in retail and e-commerce using machine learning models," *Amer. J. Eng. Technol.*, vol. 6, no. 12, pp. 163–177, Dec. 2024.
[71] A. K. Kalusivalingam, A. Sharma, N. Patel, and V. Singh, "Leveraging reinforcement learning and Bayesian optimization for enhanced dynamic pricing strategies," *Int. J. AI ML*, vol. 1, no. 3, pp. 1–14, Apr. 2020.
[72] M. A. A. Montaser, B. P. Ghosh, A. Barua, F. Karim, B. C. Das, R. E. R. Shawon, and M. S. R. Chowdhury, "Sentiment analysis of social media data: Business insights and consumer behavior trends in the USA," *Edelweiss Appl. Sci. Technol.*, vol. 9, no. 1, pp. 515–535, Jan. 2025.
[73] J. Feizabadi, "Machine learning demand forecasting and supply chain performance," *Int. J. Logistics Res. Appl.*, vol. 25, no. 2, pp. 119–142, Feb. 2022.
[74] C. Junior, P. Gusmão, J. Moreira, and A. M. M. Tome, "Time series forecasting in retail sales using LSTM and prophet," in *Handbook of Research on Applied Data Science and Artificial Intelligence in Bus. and Industry*. Hershey, PA, USA: IGI Global, 2021, pp. 241–262.
[75] J. M. Oliveira and P. Ramos, "Evaluating the effectiveness of time series transformers for demand forecasting in retail," *Mathematics*, vol. 12, no. 17, p. 2728, Aug. 2024.
[76] A. Anoop, M. Thomas, and K. Sachin, "IoT based smart warehousing using machine learning," in *Proc. Asian Conf. Innov. Technol. (ASIANCON)*, Aug. 2021, pp. 1–6.
[77] V. Pasupuleti, B. Thuraka, C. S. Kodete, and S. Malisetty, "Enhancing supply chain agility and sustainability through machine learning: Optimization techniques for logistics and inventory management," *Logistics*, vol. 8, no. 3, p. 73, Jul. 2024.
[78] G. Zheng, D. Ivanov, and A. Brintrup, "An adaptive federated learning system for information sharing in supply chains," *Int. J. Prod. Res.*, vol. 63, pp. 1–23, Jan. 2025.
[79] N. Singh, "AI and IoT: A future perspective on inventory management," *Int. J. Res. Appl. Sci. Eng. Technol.*, vol. 11, no. 11, pp. 2753–2757, Nov. 2023.
[80] A. Garcia, "Machine learning for customer segmentation and targeted marketing," in *Proc. Mach. Learn. Appl. Conf.*, vol. 3, 2023, pp. 1–15.
[81] K. Wang, T. Zhang, T. Xue, Y. Lu, and S.-G. Na, "E-commerce personalized recommendation analysis by deeply-learned clustering," *J. Vis. Commun. Image Represent.*, vol. 71, Aug. 2020, Art. no. 102735.
[82] R. S. Sucharitha and S. Lee, "GMM clustering for in-depth food accessibility pattern exploration and prediction model of food demand behavior," *Socio-Economic Planning Sci.*, vol. 83, Oct. 2022, Art. no. 101351.
[83] M. Alojail and S. Bhatia, "A novel technique for behavioral analytics using ensemble learning algorithms in e-commerce," *IEEE Access*, vol. 8, pp. 150072–150080, 2020.
[84] T. Lang and M. Rettenmeier, "Understanding consumer behavior with recurrent neural networks," in *Proc. Workshop Mach. Learn. Methods Recommender Syst.*, 2017, pp. 1–12.
[85] M. Kasimu, N. Hellen, and G. Marvin, "Explainable sentiment analysis for textile personalized marketing," in *The Fourth Industrial Revolution and Beyond: Select Proceedings of IC4IR+*. Cham, Switzerland: Springer, 2023, pp. 473–488.
[86] M. Singh, X. Hoque, D. Zeng, Y. Wang, K. Ikeda, and A. Dhall, "Do I have your attention: A large scale engagement prediction dataset and baselines," in *Proc. Int. Conf. MULTIMODAL Interact.*, Oct. 2023, pp. 174–182.
[87] K. Prasad, L. A. Xavier, S. Jain, R. Subba, S. Mittal, and N. Anute, "AI-driven chatbots for e-commerce customer support," in *Proc. Int. Conf. Adv. Comput., Commun. Appl. Informat. (ACCAI)*, May 2024, pp. 1–5.
[88] J. J. Bird and A. Lotfi, "Customer service chatbot enhancement with attention-based transfer learning," *Knowl.-Based Syst.*, vol. 301, Oct. 2024, Art. no. 112293.
[89] A. T. Imam and I. Altawaiha, "The use of the pre-trained BERT and GPT-3 models to automate the composing of use case descriptions," *Authorea Preprints*, Aug. 2023.
[90] X. Zhang and C. Guo, "Research on multimodal prediction of e-commerce customer satisfaction driven by big data," *Appl. Sci.*, vol. 14, no. 18, p. 8181, Sep. 2024.
[91] R. B. Yousif, M. G. Abd Alkreem, and A. B. Yousif, "Personalized chatbot responses using reinforcement learning and user modeling," *J. Educ. Pure Sci.*, vol. 14, no. 4, p. 462, Dec. 2024.
[92] M. G. Sivasathiya, "Emotion-aware multimedia synthesis: A generative AI framework for personalized content generation based on user sentiment analysis," in *Proc. 2nd Int. Conf. Intell. Data Commun. Technol. Internet Things (IDCIoT)*, Jan. 2024, pp. 1344–1350.
[93] S. Dhar, "BERT based sequential mining for richer contextual semantics e-commerce recommendation (BERT-SEMSRec)," M.S. thesis, School Comput. Sci., Univ. Windsor, Windsor, ON, Canada, 2024.
[94] M. Wen, D. K. Vasthimal, A. Lu, T. Wang, and A. Guo, "Building large-scale deep learning system for entity recognition in e-commerce search," in *Proc. 6th IEEE/ACM Int. Conf. Big Data Comput., Appl. Technol.*, Dec. 2019, pp. 149–154.
[95] D. Jannach and M. Ludewig, "Investigating personalized search in e-commerce," in *Proc. 30th Int. Flairs Conf.*, Jan. 2017, pp. 645–650.
[96] Z. Tang, X. Zhang, Z. Long, and X. Fu, "Multimodal neural machine translation with search engine based image retrieval," 2022, *arXiv:2208.00767*.
[97] Y. Hu, Q. Da, A. Zeng, Y. Yu, and Y. Xu, "Reinforcement learning to rank in e-commerce search engine: Formalization, analysis, and application," in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Jul. 2018, pp. 368–377.
[98] C. Han and Q. Zhang, "Optimization of supply chain efficiency management based on machine learning and neural network," *Neural Comput. Appl.*, vol. 33, no. 5, pp. 1419–1433, Mar. 2021.
[99] Y. Geng, E. Liu, R. Wang, Y. Liu, W. Rao, S. Feng, Z. Dong, Z. Fu, and Y. Chen, "Deep reinforcement learning based dynamic route planning for minimizing travel time," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, Jun. 2021, pp. 1–6.
[100] R. S. Khan, M. R. M. Sirazy, R. Das, and S. Rahman, "An AI and ML-enabled framework for proactive risk mitigation and resilience optimization in global supply chains during national emergencies," *Sage Sci. Rev. Appl. Mach. Learn.*, vol. 5, no. 2, pp. 127–144, Nov. 2022.
[101] J. Mendling, G. Decker, R. Hull, H. A. Reijers, and I. Weber, "How do machine learning, robotic process automation, and blockchains affect the human factor in business process management?" *Commun. Assoc. Inf. Syst.*, vol. 43, no. 1, p. 19, 2018.
[102] A. Z. Abideen, V. P. K. Sundram, J. Pyeman, A. K. Othman, and S. Sorooshian, "Digital twin integrated reinforced learning in supply chain and logistics," *Logistics*, vol. 5, no. 4, p. 84, Nov. 2021.
[103] T. M. Ho, K.-K. Nguyen, and M. Cheriet, "Federated deep reinforcement learning for task scheduling in heterogeneous autonomous robotic system," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 1, pp. 528–540, Jan. 2024.
[104] J. Arora, G. Kaur, M. Sethi, and S. Singh, "AI and machine learning applications for preserving privacy and data leakage of e-commerce data," in *Advances in Electronic Commerce*, 2024, pp. 59–78.
[105] A. E. Ouadrhiri and A. Abdelhadi, "Differential privacy for deep and federated learning: A survey," *IEEE Access*, vol. 10, pp. 22359–22380, 2022.
[106] S. Patel and A. Rahman, "Data privacy in the digital age: Navigating compliance and ethical challenges," *Baltic Multidisciplinary Res. Lett. J.*, vol. 1, no. 3, pp. 13–24, Nov. 2024.
[107] S. Narula, A. Afaq, S. Nagar, and M. Chaudhary, "Transformative potential and ethical challenges of generative AI in e-commerce: Data bias, algorithm bias," in *Advances in Computational Intelligence and Robotics*, 2024, pp. 317–336.
[108] M. A. Shah and P. Kumar, "Leveraging machine learning techniques to project customer behaviour through predictive analysis and ethical marketing," in *Market Grooming*, 2024, pp. 121–138.
[109] P. Urbanke, A. Uhlig, and J. J. Kranz, "A customized and interpretable deep neural network for high-dimensional business data-evidence from an e-commerce application," in *Proc. ICIS*, 2017, pp. 1–10.

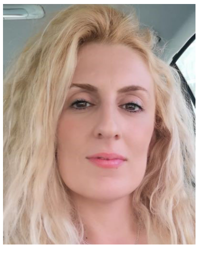
[110] S. N. Cohen, D. Snow, and L. Szpruch, "Black-box model risk in finance," in *Machine Learning and Data Sciences for Financial Markets: A Guide to Contemporary Practices*. Cambridge Univ. Press, 2023, pp. 687–717.
[111] P. Linardatos, V. Papastefanopoulos, and S. Kotsiantis, "Explainable AI: A review of machine learning interpretability methods," *Entropy*, vol. 23, no. 1, p. 18, Dec. 2020.
[112] J. Narkhede, "Comparative evaluation of post-hoc explainability methods in AI: LIME, SHAP, and grad-CAM," in *Proc. 4th Int. Conf. Sustain. Expert Syst. (ICSES)*, Oct. 2024, pp. 826–830.
[113] M. Sarkar, "Explainable AI in e-commerce: Enhancing trust and transparency in AI-driven decisions," *Innovatech Eng. J.*, vol. 2, no. 1, pp. 12–39, Jan. 2024.
[114] M. Battaglini and S. Rasmussen, "Transparency, automated decision-making processes and personal profiling," *J. Data Protection Privacy*, vol. 2, no. 4, pp. 331–349, 2019.
[115] D. Shankar, S. Narumanchi, H. A. Ananya, P. Kompalli, and K. Chaudhury, "Deep learning based large scale visual recommendation and search for e-commerce," 2017, *arXiv:1703.02344*.
[116] W. Li, M. Mikailov, and W. Chen, "Scaling the inference of digital pathology deep learning models using CPU-based high-performance computing," *IEEE Trans. Artif. Intell.*, vol. 4, no. 6, pp. 1691–1704, Dec. 2023.
[117] A. Rangra, V. K. Sehgal, and S. Shukla, "A novel approach of cloud based scheduling using deep-learning approach in e-commerce domain," *Int. J. Inf. Syst. Model. Design*, vol. 10, no. 3, pp. 59–75, Jul. 2019.
[118] X. Du, B. Bhushanam, J. Yu, D. Choudhary, T. Gao, S. Wong, L. Feng, J. Park, Y. Cao, and A. Kejariwal, "Alternate model growth and pruning for efficient training of recommendation systems," in *Proc. 20th IEEE Int. Conf. Mach. Learn. Appl. (ICMLA)*, Dec. 2021, pp. 1421–1428.
[119] M. F. Álvarez, "Edge computing security approaches and their influence on latency reduction in e-commerce payment networks," *J. Artif. Intell. Mach. Learn. Cloud Comput. Syst.*, vol. 5, no. 11, pp. 1–8, Nov. 2021.
[120] S. Kang, "Knowledge distillation approaches for accurate and efficient recommender system," 2024, *arXiv:2407.13952*.
[121] Y. Liu, J. Lu, F. Mao, and K. Tong, "The product quality risk assessment of e-commerce by machine learning algorithm on spark in big data environment," *J. Intell. Fuzzy Syst.*, vol. 37, no. 4, pp. 4705–4715, Oct. 2019.
[122] L. Caroprese, F. S. Pisani, B. M. Veloso, M. Konig, G. Manco, H. Hoos, and J. Gama, "Modelling concept drift in dynamic data streams for recommender systems," *ACM Trans. Recommender Syst.*, vol. 3, no. 2, pp. 1–28, Jun. 2025.
[123] J. Yu, M. Qiu, J. Jiang, J. Huang, S. Song, W. Chu, and H. Chen, "Modelling domain relationships for transfer learning on retrieval-based question answering systems in e-commerce," in *Proc. 11th ACM Int. Conf. Web Search Data Mining*, Feb. 2018, pp. 682–690.
[124] A. Zhao, "Deep reinforcement learning-based trading decision models in smart economic management: Cross-market forecasting and optimization," *SSRN Electron. J.*, vol. 2024, pp. 1–21, Sep. 2024.
[125] V. V. Ramasesh, A. Lewkowycz, and E. Dyer, "Effect of scale on catastrophic forgetting in neural networks," in *Proc. Int. Conf. Learn. Represent.*, 2021, pp. 1–12.
[126] G. M. van de Ven, N. Soures, and D. Kudithipudi, "Continual learning and catastrophic forgetting," 2024, *arXiv:2403.05175*.
[127] Q. Guo, Z. Li, B. An, P. Hui, J. Huang, L. Zhang, and M. Zhao, "Securing the deep fraud detector in large-scale e-commerce platform via adversarial machine learning approach," in *Proc. World Wide Web Conf.*, May 2019, pp. 616–626.
[128] M. F. Zeager, A. Sridhar, N. Fogal, S. Adams, D. E. Brown, and P. A. Beling, "Adversarial learning in credit card fraud detection," in *Proc. Syst. Inf. Eng. Design Symp. (SIEDS)*, Apr. 2017, pp. 112–116.
[129] F. Cartella, O. Anunciacao, Y. Funabiki, D. Yamaguchi, T. Akishita, and O. Elshocht, "Adversarial attacks for tabular data: Application to fraud detection and imbalanced data," 2021, *arXiv:2101.08030*.
[130] W. Hilal, S. A. Gadsden, and J. Yawney, "Financial fraud: A review of anomaly detection techniques and recent advances," *Expert Syst. Appl.*, vol. 193, May 2022, Art. no. 116429.
[131] R. Udayakumar, A. Joshi, S. S. Boomiga, and R. Sugumar, "Deep fraud net: A deep learning approach for cyber security and financial fraud detection and classification," *J. Internet Services Inf. Secur.*, vol. 13, no. 4, pp. 138–157, Dec. 2023.
[132] U. Porwal and S. Mukund, "Credit card fraud detection in e-commerce," in *Proc. 18th IEEE Int. Conf. Trust, Secur. Privacy Comput. Commun./13th IEEE Int. Conf. Big Data Sci. Eng. (TrustCom/BigDataSE)*, Aug. 2019, pp. 280–287.
[133] E. M. Al-Dahasi, R. K. Alsheikh, F. A. Khan, and G. Jeon, "Optimizing fraud detection in financial transactions with machine learning and imbalance mitigation," *Expert Syst.*, vol. 42, no. 2, p. 13682, Feb. 2025.
[134] K. H. Leung, D. Y. Mo, G. T. S. Ho, C. H. Wu, and G. Q. Huang, "Modelling near-real-time order arrival demand in e-commerce context: A machine learning predictive methodology," *Ind. Manage. Data Syst.*, vol. 120, no. 6, pp. 1149–1174, May 2020.
[135] Z. Liao, R. Zhang, S. He, D. Zeng, J. Wang, and H.-J. Kim, "Deep learning-based data storage for low latency in data center networks," *IEEE Access*, vol. 7, pp. 26411–26417, 2019.
[136] Y. Vasa, S. R. Mallreddy, and S. Jaini, "AI and deep learning synergy: Enhancing real-time observability and fraud detection in cloud environment," *Int. J. Res. Eng. Manage.*, vol. 6, no. 4, pp. 32–35, Aug. 2023.
[137] Y. Yang, L. Zhao, Y. Li, H. Zhang, J. Li, M. Zhao, X. Chen, and K. Li, "INFless: A native serverless system for low-latency, high-throughput inference," in *Proc. 27th ACM Int. Conf. Architectural Support Program. Lang. Operating Syst.*, Feb. 2022, pp. 768–781.
[138] Z. Basystiuk and Z. Rybchak, "Recommendation systems in e-commerce applications," *Inf. Syst. Netw.*, vol. 15, pp. 252–259, Jul. 2024.
[139] A. Tewari and A. Barman, "Collaborative recommendation system using dynamic content based filtering, association rule mining and opinion mining," *Int. J. Intell. Eng. Syst.*, vol. 10, no. 5, pp. 57–66, Oct. 2017.
[140] G. Dobrița, "Adaptive microservices for dynamic e-commerce: Enabling personalized experiences through machine learning and real-time adaptation," *Econ. Insights-Trends Challenges*, vol. 2023, no. 1, pp. 95–103, 2023.
[141] K. Xu, H. Zhou, H. Zheng, M. Zhu, and Q. Xin, "Intelligent classification and personalized recommendation of e-commerce products based on machine learning," 2024, *arXiv:2403.19345*.
[142] A. H. Adepoju, A. Eweje, and A. Collins, "Framework for migrating legacy systems to next-generation data architectures while ensuring seamless integration and scalability," *Int. J. Multidisciplinary Res. Growth Eval.*, vol. 5, no. 6, pp. 1462–1474, Jan. 2024.
[143] B. Celik and J. Vanschoren, "Adaptation strategies for automated machine learning on evolving data," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 9, pp. 3067–3078, Sep. 2021.
[144] E. Raj, *Engineering MLOps: Rapidly Build, Test, and Manage Production-Ready Machine Learning Life Cycles at Scale*. Birmingham, U.K.: Packt Publishing, 2021.
[145] N. Kodakandla, "Decoding MLOps: Bridging the gap between data science and operations for scalable AI systems," *Int. J. Sci. Res. Arch.*, vol. 11, no. 1, pp. 2615–2624, Jan. 2024.
[146] X.-Y. Liu, Z. Xia, H. Yang, J. Gao, D. Zha, M. Zhu, C. D. Wang, Z. Wang, and J. Guo, "Dynamic datasets and market environments for financial reinforcement learning," *Mach. Learn.*, vol. 113, no. 5, pp. 2795–2839, May 2024.
[147] L.-E. Wang, Y. Wang, Y. Bai, P. Liu, and X. Li, "POI recommendation with federated learning and privacy preserving in cross domain recommendation," in *Proc. IEEE Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)*, May 2021, pp. 1–6.
[148] S. R. Chalamala, N. K. Kummari, A. K. Singh, A. Saibewar, and K. M. Chalavadi, "Federated learning to comply with data protection regulations," *CSI Trans. ICT*, vol. 10, no. 1, pp. 47–60, Mar. 2022.
[149] J. Li, T. Cui, K. Yang, R. Yuan, L. He, and M. Li, "Demand forecasting of e-commerce enterprises based on horizontal federated learning from the perspective of sustainable development," *Sustainability*, vol. 13, no. 23, p. 13050, Nov. 2021.
[150] X. Wu, Y. Zhang, M. Shi, P. Li, R. Li, and N. N. Xiong, "An adaptive federated learning scheme with differential privacy preserving," *Future Gener. Comput. Syst.*, vol. 127, pp. 362–372, Feb. 2022.
[151] L. Shanmugam, R. Tillu, and M. Tomar, "Federated learning architecture: Design, implementation, and challenges in distributed AI systems," *J. Knowl. Learn. Sci. Technol.*, vol. 2, no. 2, pp. 371–384, Sep. 2023.
[152] X. Zhang and V. S. Sheng, "Neuro-symbolic AI: Explainability, challenges, and future trends," 2024, *arXiv:2411.04383*.
[153] M. Ansari, S. A. Ali, M. Alam, K. Chaudhary, and S. Rakshit, "Unlocking the power of explainable ai to improve customer experiences in e-commerce," in *AI-Based Data Analytics*. Boca Raton, FL, USA: Auerbach Publications, 2023, pp. 31–48.

[154] B. P. Bhuyan, A. Ramdane-Cherif, T. P. Singh, and R. Tomar, "Neuro-symbolic AI: The fusion of symbolic reasoning and machine learning," in *Neuro-Symbolic Artificial Intelligence: Bridging Logic and Learning*. Cham, Switzerland: Springer, 2024, pp. 17–27.
[155] K. Anderson, "Neurosymbolic AI revolutionizing fraud prevention systems," Tech. Rep., 2022.
[156] X. Ding, N. Seleznev, S. Kumar, C. B. Bruss, and L. Akoglu, "From detection to action: A human-in-the-loop toolkit for anomaly reasoning and management," in *Proc. 4th ACM Int. Conf. AI Finance*, Nov. 2023, pp. 279–287.
[157] Z. Lu, I. Afridi, H. J. Kang, I. Ruchkin, and X. Zheng, "Surveying neuro-symbolic approaches for reliable artificial intelligence of things," *J. Reliable Intell. Environments*, vol. 10, no. 3, pp. 257–279, Sep. 2024.
[158] R. C. Tanguturi, A. Devendran, and S. Umarani, "The role of quantum machine learning in optimizing neuromarketing insights for retail and e-commerce," in *The Quantum AI Era of Neuromarketing*. Hershey, PA, USA: IGI Global Scientific Publishing, 2025, pp. 243–254.
[159] F. Tellez and J. Ortiz, "Comparing AI algorithms for optimizing elliptic curve cryptography parameters in e-Commerce integrations: A pre-quantum analysis," *Int. J. Adv. Comput. Sci. Appl.*, vol. 15, no. 6, pp. 1539–1552, 2024.
[160] J. Shi, F. Shang, S. Zhou, X. Zhang, and G. Ping, "Applications of quantum machine learning in large-scale e-commerce recommendation systems: Enhancing efficiency and accuracy," *J. Ind. Eng. Appl. Sci.*, vol. 2, no. 4, pp. 90–103, 2024.
[161] L. M. Gutta, B. Dhamodharan, P. K. Dutta, and P. Whig, "AI-infused quantum machine learning for enhanced supply chain forecasting," in *Quantum Computing and Supply Chain Management: A New Era of Optimization*. Hershey, PA, USA: IGI Global, 2024, pp. 48–63.
[162] R. Khurana, "Applications of quantum computing in telecom e-commerce: Analysis of QKD, QAOA, and QML for data encryption, speed optimization, and AI-driven customer experience," *Quart. J. Emerg. Technol. Innov.*, vol. 7, no. 9, pp. 1–15, 2022.
[163] F. Phillipson, "Quantum computing in logistics and supply chain management an overview," 2024, *arXiv:2402.17520*.
[164] T. K. Vashishth, Vikas, K. K. Sharma, B. Kumar, S. Chaudhary, and R. Panwar, "Enhancing customer experience through AI-enabled content personalization in e-commerce marketing," in *Advances in Digital Marketing in the Era of Artificial Intelligence*. Boca Raton, FL, USA: CRC Press, 2024, pp. 7–32.
[165] H. Ko, S. Lee, Y. Park, and A. Choi, "A survey of recommendation systems: Recommendation models, techniques, and application fields," *Electronics*, vol. 11, no. 1, p. 141, Jan. 2022.
[166] H. Liu, Y. Wei, X. Song, W. Guan, Y.-F. Li, and L. Nie, "MMGRec: Multimodal generative recommendation with transformer model," 2024, *arXiv:2404.16555*.
[167] F. Pervez, M. Shoukat, M. Usama, M. Sandhu, S. Latif, and J. Qadir, "Affective computing and the road to an emotionally intelligent metaverse," *IEEE Open J. Comput. Soc.*, vol. 5, pp. 195–214, 2024.
[168] K. Bayoudh, R. Knani, F. Hamdaoui, and A. Mtibaa, "A survey on deep multimodal learning for computer vision: Advances, trends, applications, and datasets," *Vis. Comput.*, vol. 38, no. 8, pp. 2939–2970, Aug. 2022.
[169] M. Kim, W. Shin, S. Kim, and H.-W. Kim, "Predicting session conversion on e-commerce: A deep learning-based multimodal fusion approach," *Asia Pacific J. Inf. Syst.*, vol. 33, no. 3, pp. 737–767, Sep. 2023.
[170] F. Wang, Y. Zhou, S. Wang, V. Vardhanabhuti, and L. Yu, "Multi-granularity cross-modal alignment for generalized medical visual representation learning," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2022, pp. 33536–33549.
[171] S. Purushwalkam, P. Morgado, and A. Gupta, "The challenges of continuous self-supervised learning," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Jan. 2022, pp. 702–721.
[172] L. Caccia and J. Pineau, "SPeCiaL: Self-supervised pretraining for continual learning," in *Proc. Int. Workshop Continual Semi-Supervised Learn.* Cham, Switzerland: Springer, Jan. 2022, pp. 91–103.
[173] J. Yu, H. Yin, X. Xia, T. Chen, J. Li, and Z. Huang, "Self-supervised learning for recommender systems: A survey," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 1, pp. 335–355, Jan. 2024.
[174] H. Qian, Z. Dou, Y. Zhu, Y. Ma, and J.-R. Wen, "Learning implicit user profile for personalized retrieval-based chatbot," in *Proc. 30th ACM Int. Conf. Inf. Knowl. Manage.*, Oct. 2021, pp. 1467–1477.
[175] Y. Li, R. Pogodin, D. J. Sutherland, and A. Gretton, "Self-supervised learning with kernel dependence maximization," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2021, pp. 15543–15556.
[176] N. Acharya, A.-M. Sassenberg, and J. Soar, "Effects of cognitive absorption on continuous use intention of AI-driven recommender systems in e-commerce," *Foresight*, vol. 25, no. 2, pp. 194–208, Apr. 2023.
[177] E. L. Aleixo, J. G. Colonna, M. Cristo, and E. Fernandes, "Catastrophic forgetting in deep learning: A comprehensive taxonomy," 2023, *arXiv:2312.10549*.
[178] D. Muthirayan and P. P. Khargonekar, "Memory augmented neural network adaptive controllers: Performance and stability," *IEEE Trans. Autom. Control*, vol. 68, no. 2, pp. 825–838, Feb. 2023.
[179] N. Xiao and L. Zhang, "Dynamic weighted learning for unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15237–15246.
[180] Z. Zhou, M. Wang, C.-N. Yang, Z. Fu, X. Sun, and Q. M. J. Wu, "Blockchain-based decentralized reputation system in e-commerce environment," *Future Gener. Comput. Syst.*, vol. 124, pp. 155–167, Nov. 2021.
[181] D. Martinez, L. Magdalena, and A. N. Savitri, "AI and blockchain integration: Enhancing security and transparency in financial transactions," *Int. Trans. Artif. Intell.*, vol. 3, no. 1, pp. 11–20, Nov. 2024.
[182] H. Oluwabunmi Bello, C. Idemudia, and T. V. Lyelolu, "Integrating machine learning and blockchain: Conceptual frameworks for real-time fraud detection and prevention," *World J. Adv. Res. Rev.*, vol. 23, no. 1, pp. 56–68, Jul. 2024.
[183] Y. Xiao, C. Zhou, X. Guo, Y. Song, and C. Chen, "A novel decentralized e-commerce transaction system based on blockchain," *Appl. Sci.*, vol. 12, no. 12, p. 5770, Jun. 2022.
[184] E. R. Onwubuariri, B. O. Adelakun, O. P. Olaiya, and J. E. K. Ziorklui, "AI-driven risk assessment: Revolutionizing audit planning and execution," *Finance Accounting Res. J.*, vol. 6, no. 6, pp. 1069–1090, Jun. 2024.
[185] M. T. A. Tonoy, A. Islam, A. K. Das, M. K. Sah, and A. K. Mansoor, "Mitigating e-commerce security risks with blockchain: A multi-layered architecture," in *Proc. Int. Conf. Intell. Syst. Cybersecurity (ISCS)*, May 2024, pp. 1–6.
[186] B. Girimurugan, T. Venkatesan, S. Ananthavalli, S. S. S. Gogada, G. Fufa, and M. Peswani, "Blockchain for e-commerce: Revolutionizing security and trust," in *Strategies for E-Commerce Data Security: Cloud, Blockchain, AI, and Machine Learning*. Hershey, PA, USA: IGI Global, 2024, pp. 333–360.
[187] S. K. Lo, Y. Liu, Q. Lu, C. Wang, X. Xu, H.-Y. Paik, and L. Zhu, "Toward trustworthy AI: Blockchain-based architecture design for accountability and fairness of federated learning systems," *IEEE Internet Things J.*, vol. 10, no. 4, pp. 3276–3284, Feb. 2023.
[188] A. Wasilewski, "Harnessing generative AI for personalized e-commerce product descriptions: A framework and practical insights," *Comput. Standards Interface*, vol. 94, Aug. 2025, Art. no. 104012.
[189] C. Li, Z. Gan, Z. Yang, J. Yang, L. Li, L. Wang, and J. Gao, "Multimodal foundation models: From specialists to general-purpose assistants," *Found. Trends Comput. Graph. Vis.*, vol. 16, nos. 1–2, pp. 1–214, 2024.
[190] C. Herold, M. Kozielski, T. Bazazo, P. Petrushkov, S. H. Hashemi, P. Cieplicka, D. Basaj, and S. Khadivi, "Domain adaptation of foundation LLMs for e-commerce," 2025, *arXiv:2501.09706*.
[191] R. Zhang, J. He, X. Luo, D. Niyato, J. Kang, Z. Xiong, Y. Li, and B. Sikdar, "Toward democratized generative AI in next-generation mobile edge networks," *IEEE Netw.*, vol. 39, no. 3, p. 1, May 2025.
[192] S. Chandran, S. R. Syam, S. Sankaran, T. Pandey, and K. Achuthan, "From static to AI-driven detection: A comprehensive review of obfuscated malware techniques," *IEEE Access*, vol. 13, pp. 74335–74358, 2025.
[193] S. Agrawal, S. Merugu, and V. Sembium, "Enhancing e-commerce product search through reinforcement learning-powered query reformulation," in *Proc. 32nd ACM Int. Conf. Inf. Knowl. Manage.*, Oct. 2023, pp. 4488–4494.
[194] K. Balog and C. Zhai, "User simulation in the era of generative AI: User modeling, synthetic data generation, and system evaluation," 2025, *arXiv:2501.04410*.
[195] X. Zhang, F. Guo, T. Chen, L. Pan, G. Beliakov, and J. Wu, "A brief survey of machine learning and deep learning techniques for e-commerce research," *J. Theor. Appl. Electron. Commerce Res.*, vol. 18, no. 4, pp. 2188–2216, Dec. 2023.

[196] H. Jebamikyous, M. Li, Y. Suhas, and R. Kashef, "Leveraging machine learning and blockchain in e-commerce and beyond: Benefits, models, and application," in *Discover Artificial Intelligence*, vol. 3, no. 1. Springer, 2023, p. 3. [Online]. Available: https://doi.org/10.1007/s44163-022-00046-0
[197] P. Gopal and N. B. MohdNawi, "A survey on customer churn prediction using machine learning and data mining techniques in e-commerce," in *Proc. IEEE Asia–Pacific Conf. Comput. Sci. Data Eng. (CSDE)*, Dec. 2021, pp. 1–8.
[198] M. A. Al-Ebrahim, S. Bunian, and A. A. Nour, "Recent machine-learning-driven developments in e-commerce: Current challenges and future perspectives," *Engineered Sci.*, no. 1, p. 1044, Dec. 2023.
[199] S. Dheva Rajan, S. Vavilapalli, S. Hasan, R. Kumar, N. Rafa, and I. Muda, "A survey on the impact of data analytics and machine learning techniques in e-commerce," in *Proc. 5th Int. Conf. Contemp. Comput. Informat. (IC3I)*, Dec. 2022, pp. 1117–1122.
[200] C. C. Ike, A. B. Ige, S. A. Oladosu, P. A. Adepoju, O. O. Amoo, and A. I. Afolabi, "Advancing machine learning frameworks for customer retention and propensity modeling in e-commerce platforms," *GSC Adv. Res. Rev.*, vol. 14, no. 2, pp. 191–203, Feb. 2023.
[201] L. M. Policarpo, D. E. da Silveira, R. da Rosa Righi, R. A. Stoffel, C. A. da Costa, J. L. V. Barbosa, R. Scorsatto, and T. Arcot, "Machine learning through the lens of e-commerce initiatives: An up-to-date systematic literature review," *Comput. Sci. Rev.*, vol. 41, Aug. 2021, Art. no. 100414.
[202] K. Achuthan, S. Ramanathan, S. Srinivas, and R. Raman, "Advancing cybersecurity and privacy with artificial intelligence: Current trends and future research directions," *Frontiers Big Data*, vol. 7, Dec. 2024, Art. no. 1497535.
[203] M. Sharma, V. Sharma, and R. Kapoor, "Study of e-commerce and impact of machine learning in e-commerce," in *Empirical Research for E-Commerce Futuristic Systems: Foundations and Applications*. Hershey, PA, USA: IGI Globalz, 2022, pp. 1–22.
[204] M. A. Gomes and T. Meisen, "A review on customer segmentation methods for personalized customer targeting in e-commerce use cases," *Inf. Syst. e-Business Manage.*, vol. 21, no. 3, pp. 527–570, Sep. 2023.
[205] K. Achuthan, S. Sankaran, S. Roy, and R. Raman, "Integrating sustainability into cybersecurity: Insights from machine learning based topic modeling," *Discover Sustainability*, vol. 6, no. 1, p. 44, Jan. 2025.



**ELIAS DRITSAS** received the Diploma, M.Sc., and Ph.D. degrees in computer science and informatics from the Department of Computer Engineering and Informatics, University of Patras, Greece, and the M.B.A. degree from the University of Derby, U.K. His research interests focus on artificial intelligence and machine learning, big data, databases, data mining, human–computer interaction, cloud computing, security, and trust. He is the author and co-author of publications in the area of machine learning and data analysis. He participated in many development projects funded by the European Commission and Greek Secretariat of Research and Technology. He is a member of the Technical Chamber of Greece and the Association for Computing Machinery. He serves as a regular reviewer for several technical journals and conferences.



**MARIA TRIGKA** received the Diploma degree from the Department of Electrical and Computer Engineering and the master's and Ph.D. degrees in signal processing for wireless communications from the Department of Computer Engineering and Informatics, University of Patras, Greece. As a Ph.D. Candidate, she was a Research Scholar under the Project "Strengthening Human Research Potential through Doctoral Research" co-financed by Greece-State Scholarships Foundation (IKY) and European Union. Also, she has been involved in research projects co-financed by European Union and Greek national funds through the Operational Program Competitiveness, Entrepreneurship and Innovation, under the call RESEARCH–CREATE–INNOVATE that emphasized AI and ML for developing health and cultural services for the hard of hearing. She is the author and co-author of publications in the areas of signal processing, wireless communications, and machine learning. Her general research interests span the scientific areas of statistical signal processing and learning, focusing on estimation theory, adaptive/distributed signal processing, and sparse representations.